

Fuel

Multi-objective NSGA-II optimization of a solar–biogas hybrid poly-generation system based on a supercritical nitrogen Brayton cycle for simultaneous power, hydrogen, freshwater, heating, and cooling --Manuscript Draft--

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Corresponding Author:	Bhishma Karki Tribhuvan University - Tri-Chandra Multiple Campus NEPAL
First Author:	Bhishma Karki
Order of Authors:	Bhishma Karki Manal A. Abbood Hüseyin KURT Ozodbek Nematov Rustam Turakulov Bhupendra Singh Chauhan Sikata Samantaray Shahbaz Juneja Pardeep Singh Bains Chou-Yi Hsu
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Dear Editor,

I am pleased to submit our manuscript, titled " **Multi-objective NSGA-II optimization of a solar–biogas hybrid poly-generation system based on a supercritical nitrogen Brayton cycle for simultaneous power, hydrogen, freshwater, heating, and cooling**" for consideration in *Fuel*. Our study presents an innovative system that combines a hybrid solar-biogas gas turbine cycle (GTC), a nitrogen-based supercritical Brayton cycle (SBC), and a multi-effect desalination (MED) unit to enhance energy efficiency and hydrogen production while minimizing waste heat losses.

The proposed system utilizes the waste heat from the SBC and GTC to drive the MED unit for freshwater generation and an ejector-based absorption chiller for cooling. Additionally, a thermoelectric generator and an alkaline electrolyzer are integrated to further enhance waste heat recovery and hydrogen production. A comprehensive 4E (energy, exergy, exergy-economic, and environmental) analysis is conducted to assess the system's feasibility. A sensitivity analysis reveals key parameters affecting system performance, and a multi-objective optimization using the NSGA-II algorithm identifies the best trade-offs among exergy efficiency, total cost rate, and unit cost of poly-generation.

Our findings indicate that the proposed system can efficiently generate 46.77 kg/s of freshwater and 3.47 kg/h of hydrogen, with an optimized exergy efficiency of 25.10% and a reduced total cost rate of 8482.67 \$/h. The insights provided in this study are particularly relevant to the fields of sustainable desalination, hydrogen production, and renewable energy integration.

Given the journal's focus on renewable energy systems, hydrogen production, and sustainable technologies, we believe our manuscript aligns well with its scope. We are confident it will be of interest to researchers and professionals working in the areas of renewable energy integration, hydrogen production, and water desalination. This manuscript is original, has not been published elsewhere, and is not under consideration by another journal. All authors have approved the submission, and there are no conflicts of interest to declare.

Thank you for considering our manuscript for publication in *Fuel*. We look forward to your feedback and the opportunity to contribute to your esteemed journal.

Sincerely,

Dr. Bhishma Karki

Executive Chairman at National Research Council Nepal

Highlights:

- A novel poly-generation system using a hybrid heat source.
- The topping system is a gas turbine cycle with a double-stage compressor.
- Six subsystems are used as bottoming layouts.
- Energy and exergy-economic analyses are conducted through a parametric study.
- The multi-objective optimization results in an exergy efficiency of 25.1% and a unit cost of 19.01 \$/GJ.

Multi-objective NSGA-II optimization of a solar–biogas hybrid poly-generation system based on a supercritical nitrogen Brayton cycle for simultaneous power, hydrogen, freshwater, heating, and cooling

Bhishma Karki^{1,2,3*}, Manal A. Abbood^{4}, Hüseyin KURT⁵, Ozodbek Nematov⁶, Rustam Turakulov⁷, Bhupendra Singh Chauhan⁸, Sikata Samantaray⁹, Shahbaz Juneja¹⁰, Pardeep Singh Bains^{11,12}, Chou-Yi Hsu¹³**

1. Department of Physics, Tri-Chandra Multiple Campus, Tribhuvan University, Kathmandu 44600, Nepal

2. National Research Council Nepal, New Baneshwor-10, Kathmandu 44600, Nepal

3. Department of Physics and Research Centre, Tuljaram Chaturchand College, Baramati, Pune 413102, India

4. Applied Science Department, University of Technology, Baghdad, Iraq

5. Department of Electrical and Electronics Engineering, Faculty of Engineering, Istanbul Aydin University, Istanbul, Türkiye

6. Jizzakh State Pedagogical General History departemnt, Jizzakh University, Jizzakh, Uzbekistan

7. Tashkent State Medical University, Tashkent 100109, Uzbekistan

8. Department of Mechanical Engineering, Institute of Engineering and Technology, GLA University, Mathura, India

9. Department of Mechanical Engineering, Siksha 'O' Anusandhan (Deemed to be University), Bhubaneswar, Odisha-751030, India

10. Department of Mechanical Engineering, Chandigarh University, Mohali, Punjab, India

11. Centre for Research Impact and Outcome, Chitkara University, Rajpura, Punjab, India

12. Department of Mechanical Engineering, Sharda School of Engineering & Sciences, Sharda University, Greater Noida, India

13. Department of Pharmacy, Chia Nan University of Pharmacy and Science, Tainan 71710, Taiwan

***Corresponding Author: Bhishma Karki (bhishma.karki@trc.tu.edu.np)**

****Corresponding Author: Manal A. Abbood (manal.a.abbood@uotechnology.edu.iq)**

Abstract

This study focuses on the design, performance assessment, and optimization of an integrated poly-generation configuration that combines a solar-assisted, biogas-fueled gas turbine cycle with a nitrogen-based supercritical Brayton cycle and a multi-effect desalination unit. Solar energy together with biogas serves as the primary energy supply to the gas turbine, while the high-temperature exhaust streams are efficiently utilized through the supercritical Brayton cycle and desalination subsystem. Further recovery of waste heat is accomplished by incorporating a thermoelectric generator alongside an ejector-driven $\text{NH}_3\text{-LiNO}_3$ absorption refrigeration system. The electricity produced by the supercritical Brayton cycle is supplied to an alkaline electrolyzer for hydrogen generation. The proposed system is analyzed using a comprehensive 4E framework, including energy, exergy, exergoeconomic, and environmental evaluations, with particular attention to CO_2 emissions and their mitigation. Under design conditions, the results indicate that the system is capable of simultaneously producing electricity, hydrogen, freshwater, heating, and cooling, while improving overall resource utilization and environmental performance. Parametric analysis shows that increasing heliostat field area, raising the solar receiver inlet temperature, and improving the methane fraction in biogas positively influence thermo-economic performance and CO_2 -related indicators, whereas higher gas turbine inlet temperatures negatively affect system performance. A multi-objective optimization approach based on the NSGA-II algorithm is employed to concurrently enhance efficiency, minimize costs, and reduce environmental impacts. The optimized configuration yields an exergy efficiency of 25.10%, a total cost rate of \$8,482.67 h^{-1} , and a poly-generation exergy unit cost of 19.01 \$/GJ. Finally, the performance of the proposed system is benchmarked against recently reported solar-assisted hybrid multi-generation systems, confirming its competitiveness and robustness.

Keywords: Gas turbine cycle; Waste heat recovery; Multi-effect desalination, Alkaline electrolyzer; Exergy-economic; Optimization

1. Introduction

The hybrid solar power tower (SPT) and biogas combustion chamber (CC) provide a sustainable energy source for gas turbine cycles (GTCs), enhancing efficiency and reducing greenhouse gas emissions. The SPT utilizes concentrated solar energy to produce high-temperature heat, while the biogas CC ensures continuous operation, compensating for solar intermittence [1, 2]. This hybrid approach leverages renewable energy and organic waste-derived fuel to achieve a lower carbon footprint. Waste energy recovery in GTC is essential to improve overall system efficiency by capturing and reusing heat from exhaust gases. In addition to minimizing energy losses, this promotes the circular economy, lessens reliance on fossil fuels, and improves the energy systems' sustainability both environmentally and economically [3, 4]. Mohammadi et al. [5] proposed two innovative hybrid power generation systems integrating SPT and gas turbine technologies, utilizing supercritical CO_2 (s- CO_2) cycles as bottoming cycles. These systems harness waste heat from a gas turbine, which is partially powered by a SPT capable of generating thermal energy at temperatures reaching up to 1223 K. A configuration utilizing a recuperative cycle with s- CO_2 recompression achieved a competitive levelized cost of electricity (LCOE) of \$83.16 MWh. However, thermodynamic

and economic analyses revealed that the cycle's efficiency was measured at 0.4608. Zhang et al. [6] introduced a novel approach by integrating a GTC with a solar-powered methanol decomposition reaction system to optimize the utilization of both methanol and solar energy. Their findings demonstrated that the incorporation of solar energy significantly increased the methanol production efficiency, elevating the yield from 3.17 kWh/kg to 3.75 kWh/kg. Zoghi et al. [7] examined a trigeneration system utilizing an GTC and air bottoming cycle, which is fueled by a combination of natural gas and SPT. Their study demonstrated that the integration of an organic Rankine cycle (ORC), a LiBr-H₂O absorption chiller (ACH), and a heat recovery steam generator (HRSG) enhanced the net power output by 8.9%, while simultaneously improving energy efficiency and exergy efficiency by 13.39 and 4.43 percentage points, respectively. The subsystems increased the overall cost rate by 7.26% while producing 36.9 MW of power, 10.6 MW of heating, and 8.8 MW of cooling. In order to provide multipurpose outputs including electricity, hydrogen, freshwater, and heating, Alirahmi et al. [8] studied a unique energy system that combined biomass with solar energy. The system integrated a gasifier, a heliostat field, and a waste-heat recovery unit. It incorporated two gas turbines, a steam Rankine cycle (SRC), and an ORC, achieving an optimal total cost rate of \$224.1 per hour and an exergy efficiency of 26.7%. To enhance system performance, Izadi et al. [9] conducted a study on a solar-augmented gas turbine power plant integrated with thermoelectric generators (TEGs), a LiBr-H₂O ACH, a SRC, an ORC, and a single-flash desalination unit. Based on their findings, the integration of solar energy resulted in a reduction of fuel consumption from 0.371 kg/s to 0.1839 kg/s, while enhancing energy efficiency by 2.92%. The integration of waste heat recovery in a hybrid solar-biomass system powered by a regenerative GTC and recompression supercritical CO₂ Brayton cycle was examined by Cao et al. [10]. Energy efficiency increased by 22.48% in hybrid mode and 29.6% in biomass-only mode, according to their analysis, while exergy efficiency climbed by 6.18% and 7.6%.

The supercritical Brayton cycle (SBC) has emerged as a highly efficient technology for waste energy recovery due to its superior thermal performance and compact design. Its ability to operate at high pressures and temperatures allows for effective utilization of low-grade heat sources, making it ideal for industrial waste heat recovery applications [11, 12]. Moreover, the cycle's flexibility to use various supercritical working fluids, such as carbon dioxide, nitrogen, and helium, enables optimization for specific waste heat conditions, further enhancing its efficiency and adaptability. This versatility makes the supercritical Brayton cycle a promising solution for sustainable energy systems [13, 14]. For tiny modular sodium-cooled fast reactors, Olumayegun et al. [15] conducted a thermodynamic performance analysis of a novel dual-shaft nitrogen-based closed Brayton cycle. According to their research, the suggested cycle reduced space and simplified the design of the turbomachinery while achieving an efficiency of 39.44% that was equivalent to the reference single-shaft cycle. In contrast to s-CO₂, Coco-Enriquez et al. [16] investigated different supercritical fluids, such as nitrogen, ethane, methane, and xenon, for Brayton power cycles in conjunction with line-focusing solar power plants. Nitrogen achieved the highest gross efficiency of 45.8% under wet-cooling conditions (compressor inlet temperature (CIT) = 32°C), surpassing s-CO₂ at 44.37%, while maintaining stable performance across varying turbine inlet pressures (TIP). Ethane and Methane showed higher heat transfer coefficients but lower efficiencies at 40.74% and 32.15%, respectively. A methodical, multi-step approach was presented by Javanshir et al. [17] to determine the best working fluids for regenerative and basic SBCs in concentrated solar power plants. Their analysis revealed that Nitrogen and CO₂ offer the highest thermal efficiencies under varying operating conditions, while Helium consistently

provides the best net specific work output. Sarkar [18] conducted thermodynamic analyses and optimizations of the N₂O recompression Brayton cycle, comparing its performance with the CO₂ counterpart. Their results revealed that N₂O achieves superior thermal efficiency, with a maximum deviation of 1.2% higher than CO₂, and better second-law efficiency under similar operating conditions. Khanmohammadi et al. [19] analyzed an innovative solar tower system integrating a helium Brayton cycle (HBC), TEG, and reverse osmosis (RO) unit, achieving enhanced performance metrics. The proposed GT-HBS/TEG-RO system attained an energy efficiency of 62.44%, significantly higher than the 43.88% of the primary GT-HBS system, with a 20.23% improvement in exergy efficiency.

A multi-effect desalination (MED) system offers significant advantages for freshwater production, particularly when powered by waste heat. By utilizing multiple stages of evaporation and condensation, MED achieves higher thermal efficiency and lower energy consumption compared to conventional desalination methods like reverse osmosis or single-effect distillation [20, 21]. Its ability to integrate seamlessly with waste heat sources from industrial processes or power plants enhances overall system efficiency while reducing operational costs and environmental impact. Unlike reverse osmosis, which relies heavily on electricity and high-pressure pumps, MED systems capitalize on low-grade thermal energy, making them ideal for sustainable and cost-effective desalination [22, 23]. A novel multi-generation plant powered by a central receiver-based concentrated solar system was presented in this part by Behzadi et al. [24], who integrated a MED unit to reduce energy losses and improve system efficiency. Their analysis showed that the MED unit, coupled with a Rankine cycle and flue gas condensation, achieved a 53.2% reduction in exergy loss, contributing significantly to overall system performance. A brand-new solar tower-based multigeneration system was created by Colakoglu and Durmayaz [25]. The system demonstrated energy and exergy efficiencies of 51.99% and 37.99%, respectively, and generated notable reductions in CO₂ emissions of 513.8 kg/h and a unit product profit of \$0.0352/kWh. The system's financial feasibility was demonstrated by its economic study, which showed a discounted payback period of 4.79 years. To improve energy recovery and reduce environmental effect, Fakhari et al. [26] created a tri-generation system that includes parabolic trough solar collectors and a triple-pressure HRSG. While the cooling load reached 5.98 MW, a MED unit powered by the system's low-pressure steam produced 181.82 m³/h of freshwater. According to optimization results, CO₂ emissions decreased from 6.31 kg/s to 6.01 kg/s, and energy efficiency increased by 20.17%. Elsayed et al. [27] designed an integrated multigeneration system that combines a gas turbine, a supercritical CO₂ recompression Brayton cycle, and a MED unit equipped with a thermal vapor compressor to simultaneously generate power, cooling, and freshwater. Their proposed system demonstrated a significant reduction in CO₂ emissions by 39.3% and achieved a 34.2% decrease in energy consumption relative to conventional standalone systems. The MED system produced 650 kg/s of freshwater, while the overall system exhibited a 57.5% energy utilization factor and a 39.7% exergy efficiency. Hai et al. [28] introduced an advanced multi-generation system that combines biomass digesters and gasifiers with a s-CO₂ cycle and a MED unit. This integrated system was subjected to comprehensive assessments encompassing exergy, energy, economic, and environmental analyses. According to their research, the digester performed better than the gasifier in terms of energy efficiency, net output power, and freshwater generation, but the exergy destruction in the digestion unit (11,337 kW) was more than in the gasification unit (9,629 kW). Morid and Manesh [29] explored the integration of GTC with hybrid multi-effect distillation adsorption desalination (MEDAD) systems in a novel poly-generation setup, addressing energy, exergy, and

environmental aspects. Their findings highlighted that optimizing MEDAD stages enhanced freshwater production by 95.25% while reducing fuel consumption by 22%. The system demonstrated a net power enhancement of 5% for the GTC, achieving overall energy efficiencies of 34.27% in single-generation mode and 65.65% in multi-generation mode.

The hybrid solar power tower-biogas combustion chamber GTC is a novel arrangement that is presented and examined in this paper as a topping system. The residual energy from the topping GTC is subsequently harnessed by a nitrogen-based SBC and a HRSG within a MED system. Additionally, the waste heat generated during the compressor stages is recuperated by a hot water unit (HWU) integrated into the GTC. Subsequently, the waste energy from the SBC is harnessed through the integration of a TEG and an ejector-based ammonia-lithium nitrate ($\text{NH}_3\text{-LiNO}_3$) ACH. Furthermore, the combined output power generated by the SBC and TEG is directed to an alkaline electrolyzer (AE) for hydrogen production. A distinctive aspect of the present study is the innovative utilization of nitrogen within the SBC and, ejector along with the $\text{NH}_3\text{-LiNO}_3$ working fluid in the ACH. This novel approach enhances the overall efficiency of the poly-generation system and represents a significant advancement, as it has not been explored in previous research. conducting a thorough evaluation of the system under investigation, encompassing energy, exergy, exergo-economic, and environmental (4E) analyses, the subsequent phase involves parametric analysis and optimization of the design. Ultimately, the findings of this study are benchmarked against previous solar-integrated fuel-based GTC waste heat recovery systems to elucidate the relative merits and limitations of the proposed configuration.

2. System description

Fig. 1 presents a conceptual illustration of the present investigation, depicting a proposed poly-generation system. Compressors 1 and 2 are used in this configuration by the GTC to increase the ambient air pressure at point 1 with an equal pressure ratio. To optimize system performance, an intercooler in the form of a heat exchanger is incorporated between the compressors. Subsequently, the temperature of the air at point 4 is elevated in three distinct phases. The recuperator (REC) initially uses the turbine's exhaust gas to reheat the 4 to 5 stream. The air temperature is then increased using a combustion chamber (CC) powered by biogas and a SPT. The high-pressure and high-temperature fluid stream (point 7) ultimately facilitates power generation within the topping cycle through the GTC turbine. The gas, which is depleted, transfers its energy within the REC (point 8) and SBC RHX (point 9) components before being conveyed to the HRSG at point 10. The SBC system involves a dual-phase cooling process of the Nitrogen stream, which is accomplished through the utilization of $\text{NH}_3\text{-LiNO}_3$ ACH and TEG subsystems to provide cooling and additional power, respectively. Nitrogen is used as working fluids in the SBC. The compressed Nitrogen is introduced into the SBC compressor at point 16-a, where it undergoes a pressure increase and consequent temperature rise. Subsequently, the compressor outlet fluid is preheated in the REC (points 17 to 18) before entering the RHX (points 18 to 13). The Nitrogen is then heated by the RHX process before entering the turbine. The SBC turbine derives its power from the heated Nitrogen flow at point 13. At stage 10, waste heat is employed as the principal thermal energy source for the MED subsystem. The MED subsystem uses the thermal energy of steam 20 as its input heat. This heat then exits the system at state 21 and is reinjected into the HRSG to produce freshwater through the first stage of the unit. In the initial phase of this system, feed water enters and absorbs the thermal energy of stream 20 before

174 evaporating. As a result, brine and desalinated steam are created, and the second stage recovers the latent heat of the
175 desalinated steam. This process is aimed at ensuring efficient use of thermal energy and subsequently generating
176 freshwater. In accordance with the process, stage n-1 generates latent heat, which is used for desalination of steam,
177 and subsequently, stage n produces the final desalinated steam. Through a condenser, the incoming saltwater then
178 rejects the latent heat of this steam. At stage 22, the saltwater is sent to the condenser, where it absorbs heat. When it
179 exits, the stream is divided into feed water and cooling water (state 25). Brine is rejected at stage 24, freshwater is
180 generated at state 23, and cooling water is returned to the sea. In this process, the latent heat is effectively harnessed
181 to generate freshwater. Subsequently, within the bottoming system, the combined output power from the SBC turbine
182 and TEG is directed to an AE for the production of hydrogen at point 46 and oxygen at point 47.

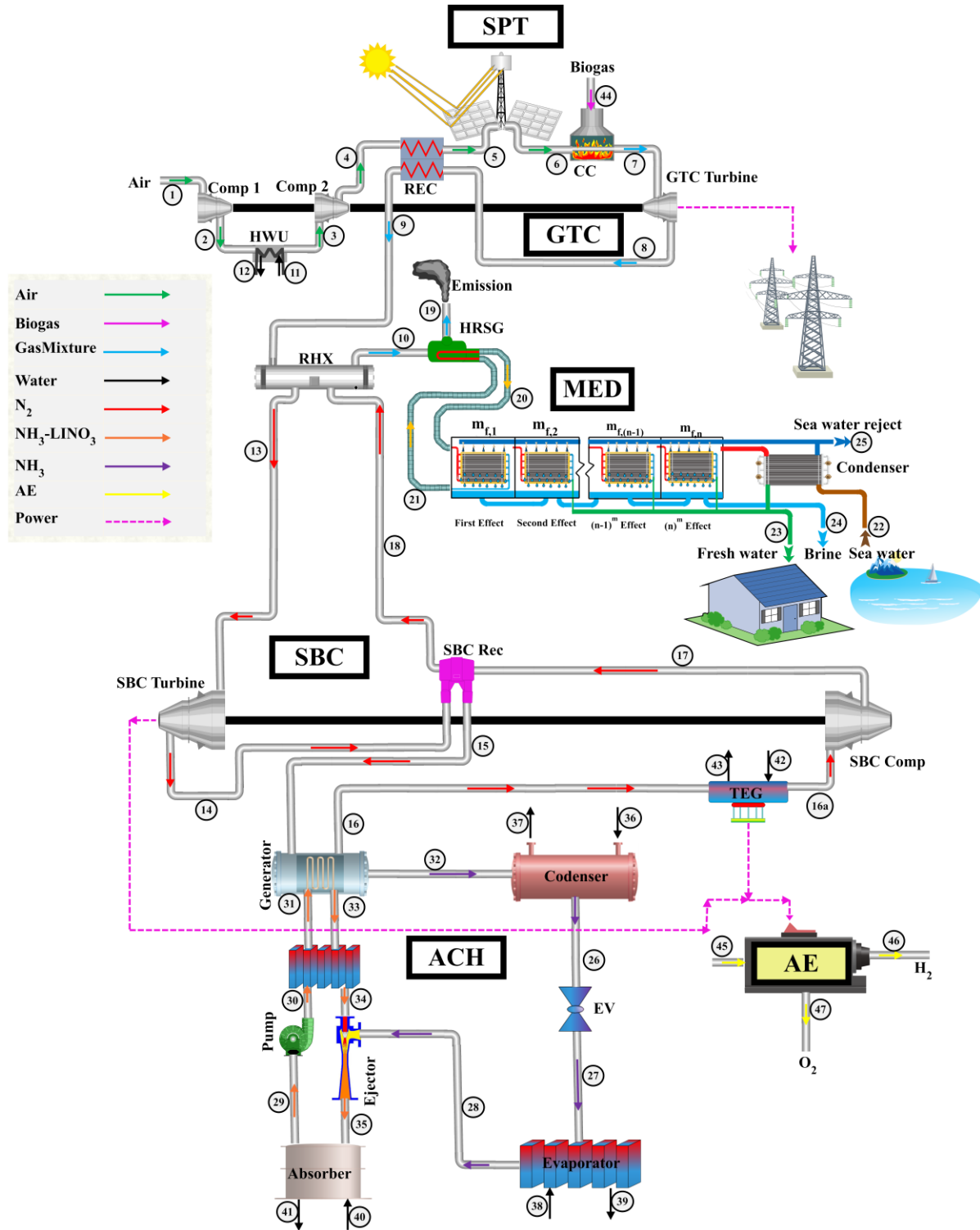


Fig. 1. diagram showing the recommended arrangement

3. Methodology

3.1. Mathematical modeling

The present study conducts a comprehensive analysis of the energy, exergy, exergy-economic, and environmental performance of the proposed configuration by developing relevant correlations within the engineering equation solver (EES) software. Subsequently, the output data generated by the EES program is integrated with the MATLAB NSGA-II toolbox through the application of artificial neural network (ANN) modeling. Certain simplifying assumptions were incorporated during the simulation process, as outlined in the subsequent sections.

- In the simulations of the whole system [30-34]:
 - In the framework of steady-state circumstances, the conversion equations have been developed.
 - The REC, SPT, and CC exhibit a 5% pressure drop within the air stream, whereas the REC and RHX demonstrate a 3% pressure reduction in the gas stream. Conversely, other heat exchangers and interconnecting pipelines experience a negligible decline in pressure.
 - All of the pipes and components that interconnect are adequately insulated.
 - The kinetic and potential constituents of the energy and exergy formulations have been omitted.
 - All turbo-machines are assumed to have constant isentropic efficiencies.
 - The relationships and principles applicable to ideal gas mixtures are employed to determine the corresponding thermodynamic properties and parameters of the gas mixture.
 - The water produced by the MED unit through desalination process is devoid of any salt content.
 - The total feed flow rate is uniformly distributed among all effects in an equitable manner.
 - The presumed notion is that the temperature discrepancy amongst the flows in all effects is equivalent.
- In the simulations of the ejector in ACH [35, 36]:
 - There is an absence of heat transfer from sources outside of the system.
 - The primary and secondary fluids experience stagnation conditions upon entering the ejector.
 - It is observed that the diluted solution flows through the nozzle from the high-pressure region to the evaporator's lower-pressure zone.
 - Efficiencies are in charge of accounting for the existence of non-idealities, such as losses and frictional forces.
 - Complete mixing occurs at the mixing section's exit due to the fluid characteristics' consistency over the cross-sectional region.
 - Potential energy has a hugely negligible magnitude.
 - Due to the exceedingly small Mach number exhibited by ammonia vapor within the ejector, it is deemed appropriate to consider the flow as incompressible.
 - It should be noted that the mixing tube and the diffuser do not undergo any absorption processes.

Appendix A provides the fundamental mathematical expressions required for the simulation of SPT, CC, ejector, TEG, MED and AE. The schematic diagram of the ejector, illustrated in Fig. 2, is essential for interpreting the equations presented in Appendix A.

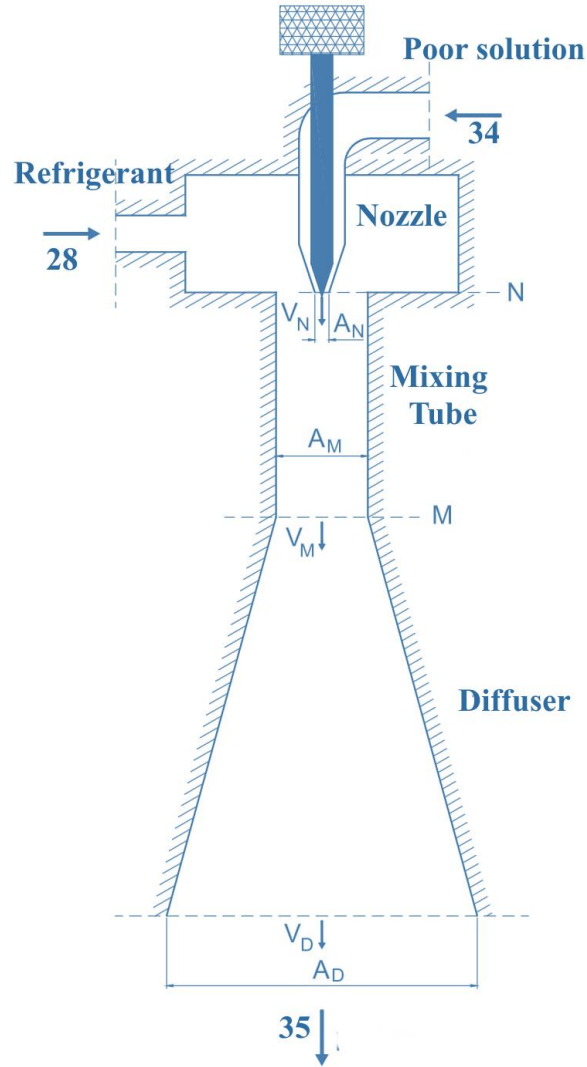


Fig. 2. Diagrammatic illustration of the ejector utilized at the absorber's entry [35]

3.2. Thermodynamic assessment and exergy-based economic analysis

Below are the mass, energy, exergy, and exergy-economic formulae needed to simulate the whole arrangement [37, 38]:

$$\sum \dot{m}_i = \sum \dot{m}_e \quad (1)$$

$$\dot{Q} + \sum \dot{m}_i h_i = \dot{W} + \sum \dot{m}_e h_e \quad (2)$$

$$\dot{E}x_Q + \sum \dot{m}_i ex_i = \dot{E}x_w + \sum \dot{m}_e ex_e + \dot{E}x_{\text{dest}} \quad (3)$$

$$c_Q \dot{E}x_Q + \sum c_i \dot{E}x_i + \dot{Z} = \sum c_e \dot{E}x_e + c_w \dot{W} \quad (4)$$

with reference to Ref. [37], Table S1 in Supplementary Material provides a detailed description of the energy balance and fuel and product exergies in the various layout apparatuses. A comprehensive overview of the exergy-economic analyses and their corresponding supplementary interconnections across various configurations is presented in Table S2 in Supplementary Material. Additionally, Appendix B incorporates the initial procurement costs of the constituent components.

3.3. Output parameters

The net output power and exergy efficiency are determined through the application of the following equations:

$$\dot{W}_{\text{net}} = \dot{W}_{GTC} = \dot{W}_{GTC,tur} - \dot{W}_{com1} - \dot{W}_{com2} \quad (5)$$

$$\eta_{ex,poly} = \frac{\dot{W}_{\text{net}} + \dot{E}x_{p,HWU} + \dot{E}x_{p,ACH,eva} + \dot{E}x_{23} + \dot{E}x_{46}}{\dot{E}x_{f,SPT} + \dot{E}x_{44}} \quad (6)$$

To conduct an exergy-economic evaluation, the total cost rate and the unit cost of poly-generation are subsequently determined as follows:

$$\dot{C}_{\text{tot}} = \sum \dot{Z}_k + \sum \dot{C}_{\text{dest},k} + \dot{C}_{\text{fuel}} + \dot{C}_{\text{env}} \quad (7)$$

$$c_{\text{poly}} = \frac{\dot{C}_{W,\text{net}} + \dot{C}_{12} + \dot{C}_{39} + \dot{C}_{23} + \dot{C}_{46}}{\dot{W}_{\text{net}} + \dot{E}x_{12} + \dot{E}x_{39} + \dot{E}x_{23} + \dot{E}x_{46}} \quad (8)$$

The following formula will be used to calculate the rate of cost for the biogas fuel used and the rate of environmental cost [39]:

$$\dot{C}_{\text{fuel}} = c_{44} \dot{m}_{44} \text{LHV}_{\text{biogas}} \quad (9)$$

$$\dot{C}_{\text{env}} = 0.024 \dot{m}_{\text{CO}_2} \quad (10)$$

3.4. Optimization

It is important to note that energy conversion systems often involve multiple competing objectives that must be simultaneously addressed. Generally, the financial investment required to develop an efficient system is lower than that needed to scale up the system while maintaining efficiency. When dealing with multiple objectives, a set of optimal solutions can often be identified. In the present study, a MATLAB-based sub-program known as NSGA-II was utilized to determine a set of optimal points. As highlighted in the parametric analysis in Subsection 4.2, three output parameters—namely, poly-generation exergy efficiency, total exergy cost rate, and unit cost of outputs—exhibit

conflicting trends. As a result, these parameters can be framed as a three-objective optimization problem. It is essential to emphasize that while minimizing the unit cost of products and the total cost rate is critical, maximizing exergy efficiency is equally significant.

$$Objfun_1 = \max(\eta_{ex,poly}) \quad (11)$$

$$Objfun_2 = \min(\dot{C}_{tot}) \quad (12)$$

$$Objfun_3 = \min(c_{poly}) \quad (13)$$

The range of the design variables being considered has been detailed in [Table 1](#).

Table 1 The variation of inputs exhibits a range within the domain of optimization

Parameter	Range
R_p	4 – 11
$T_5(^{\circ}\text{C})$	300-410
x_{ch_4}	0.5-0.7
$GTIT(^{\circ}\text{C})$	960-1100
N_{mirror}	600 – 1000

A point from the derived Pareto front that demonstrates the minimal euclidean distance relative to the origin is selected to represent the optimal system performance. The objective functions are normalized by considering the minimum and maximum parameter values within the defined range of variation to identify this optimal point:

$$objfun_{norm} = \frac{objfun - objfun_{min}}{objfun_{max} - objfun_{min}} \quad (14)$$

3.5. Validation

The entirety of the current investigation has been replicated within the EES software. Initially, the resulting data pertaining to GTC, the $\text{NH}_3\text{-LiNO}_3$ ACH with ejector numerical output, MED, and ACH COP and AE have been juxtaposed with the findings from prior related research, as depicted in [Tables 2-4](#) and [Figs. 3](#) and [4](#). All input data have been utilized in accordance with the aforementioned references. Owing to the negligible disparities between the output parameters, the current simulation appears to be accurate. It should be mentioned that the figures in [Tables 2](#) and [3](#) match those in [Figs. 1](#) and [2](#) of Refs. [\[40\]](#) and [\[35\]](#) respectively.

Table 2 Assessment of GTC simulation accuracy using biogas fuel

State point	P (bar)		T (K)		$\dot{m}$ (kg s^{-1})	
	Result	Ref. [40]	Result	Ref. [40]	Result	Ref. [40]
1	1.013	1.013	298.15	298.15	4.77	4.77
2	10.23	10.23	612.65	612.6	4.77	4.77
3	9.72	9.72	650	650	4.77	4.77

4	9.23	9.23	1250	1250	4.96	4.92
5	1.12	1.12	822.65	819.9	4.96	4.92
6	10.7	10.7	789.15	782.9	4.96	4.92
Input values: CPR=10.1, $T_3=377$ °C, $T_4=977$ °C, $\dot{m}_{air} = 4.77 \text{ kg s}^{-1}$						

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Table 3 Validation NH₃ -LiNO₃ ACH with ejector modeling

State point	T (°C)		$\dot{m} \text{ (kg s}^{-1}\text{)}$		Ammonia concentration (%)		Specific enthalpy ($\text{kJ kg}^{-1} \text{ K}^{-1}$)	
	Ref.	Result	Ref. [35]	Result	Ref. [35]	Result	Ref. [35]	Result
1	35	35	0.27	0.2737	1	1	366.08	366
2	0	0	0.27	0.2737	1	1	366.08	366
3	0	0	0.27	0.2737	1	1	1462.23	1462
4	35	35	2.17	2.043	0.51	0.5109	-3.75	-3.213
5	35.3	35.31	2.17	2.043	0.51	0.5109	-2.81	-2.242
6	66.41	66.05	2.17	2.043	0.51	0.5109	96.3	95.95
14	90	90	0.27	0.2737	1	1	1646.23	1646
15	90	90	1.89	1.769	0.44	0.4353	149.62	149.1
16	51.81	51.72	1.89	1.769	0.44	0.4353	36.18	35.73
17	46.17	45.98	2.17	2.043	0.44	0.377	216.3	226.8
18	100	100	0.23	0.2297	-	-	2675.3	2676
19	100	100	0.23	0.2297	-	-	419.06	419.1
20	27	27	16.75	16.75	-	-	113.12	113.1
21	32	32	16.75	16.75	-	-	134.04	134
22	8	7.9	14.29	14.58	-	-	33.61	33.19
23	3	3	14.29	14.58	-	-	12.61	12.61
24	27	27	22.79	22.47	-	-	113.12	113.1
25	32	32	22.79	22.47	-	-	134.04	134

276

277

Table 4 The validation of MED with Abdelhay et al. [33]

Plant condition	Abdelhay et al. [33]	Present study
Operation parameters		
Heating steam Temperature (°C)	69	69
No. of Effects	8	8
Seawater Temperature (°C)	25	25

Seawater Salinity (ppm)	35000	35000
Brine Salinity (ppm)	70000	70000
Top Brine Temperature (°C)	38.77	38.77
Plant Performance		
Gain output ratio	5.42	5.43
Distillated water (kg s^{-1})	6.36	6.37
Heat Transfer Area (m^2)	1861.81	1869.1

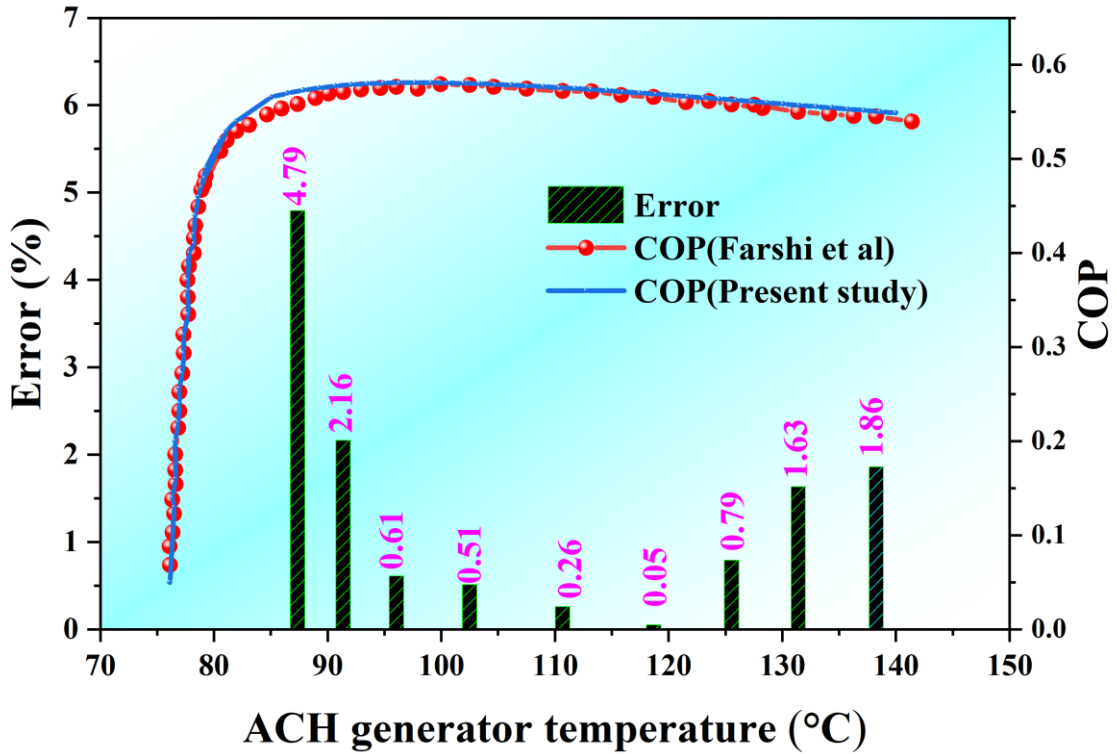


Fig. 3. Farshi et al.'s confirmation of COP [35] in relation to generator temperature

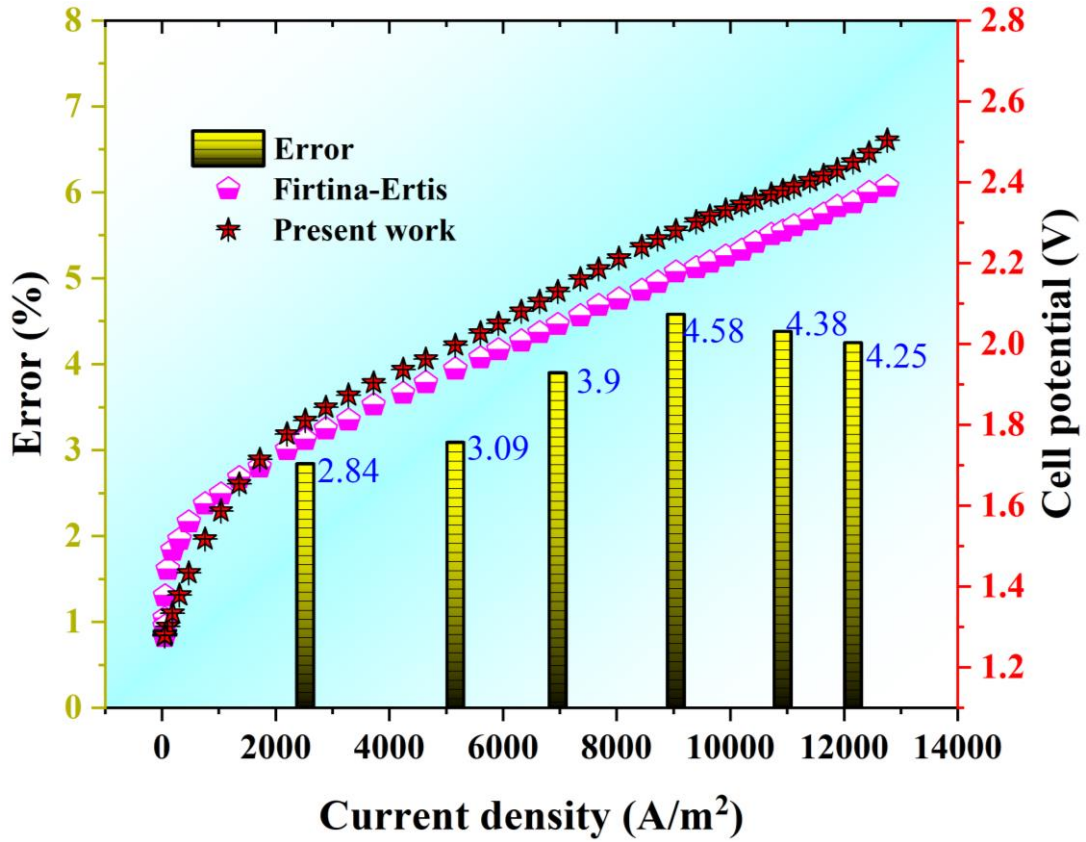


Fig. 4. Firtina-Ertis [41] was used to validate the AE voltage

Furthermore, it is demonstrated in Fig. 5 that the outputs generated by the ANN have been successfully confirmed. The R^2 metric, which depicts the deviation between the anticipated and factual outputs, indicates that the highest level of accuracy is achieved when the value of y equals x. In light of this metric, the evaluations for exergy efficiency, poly-generation unit costs and overall cost rate have been deemed satisfactory.

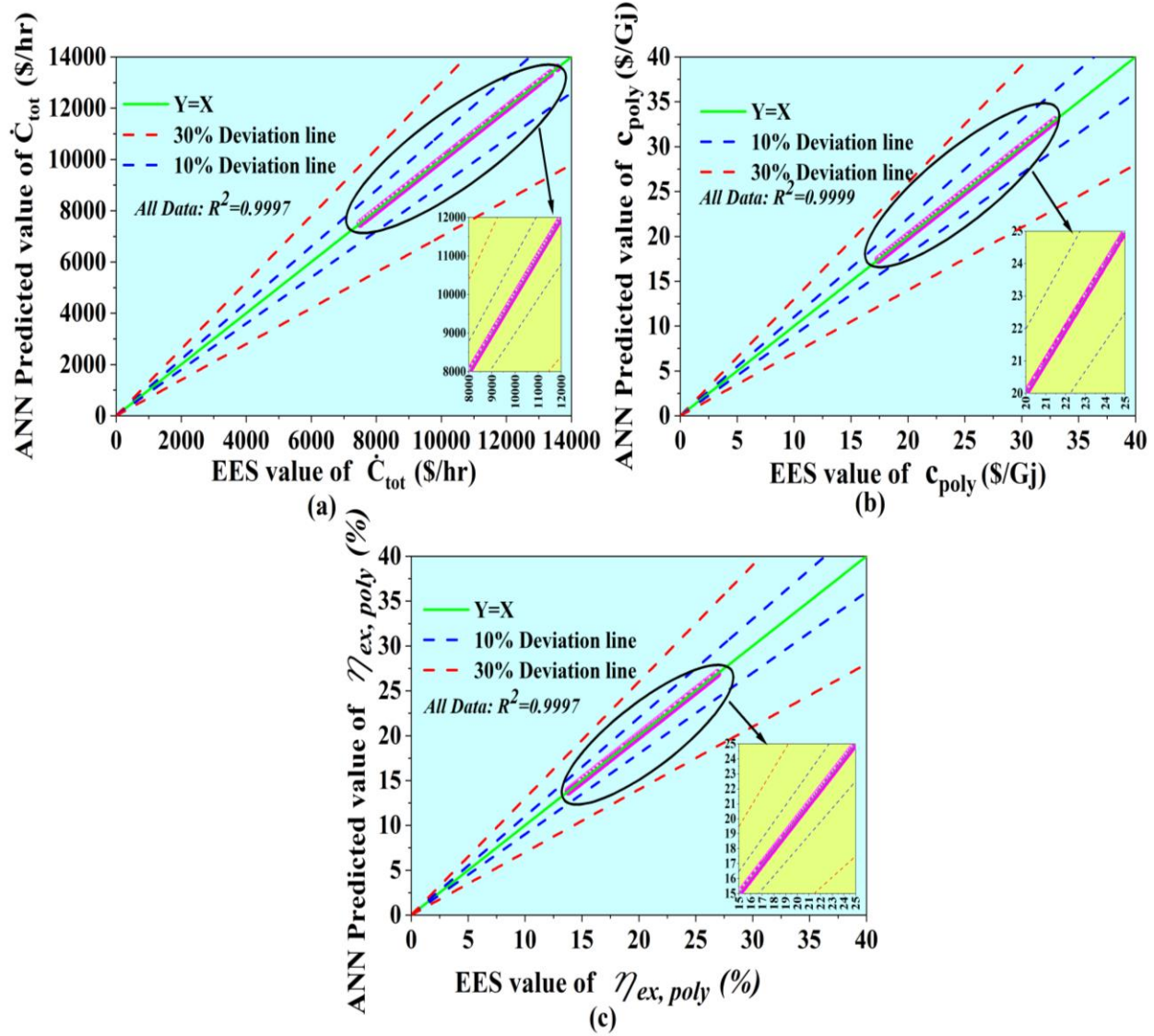


Fig. 5. ANN model output accuracy validation using EES values for (a) c_{poly} , (b) $\eta_{ex, poly}$, (c) $\dot{C}_{tot}$

4. Result and discussion

4.1. Initial scenario outcomes

Table 5 outlines the essential input parameters necessary for conducting the fundamental analysis of the system. Furthermore, Fig. 6 illustrates the sequential procedures involved in the simulation and optimization processes.

Table 5 Provide input parameters for the baseline scenario

Parameter	Value
T_0	25 °C

P_0	101.3 kPa
GTC [40, 42]	
$\dot{m}_{\text{air}}$	100 kg s ⁻¹
$R_{p,com}$	10
T_3	30 °C
T_5	400 °C
x_{ch_4}	0.6
GTIT	1000 °C
$\eta_{isen,com}$	0.87
$\eta_{isen,GT}$	0.89
SPT [43, 44]	
N_{mirror}	800
η_{field}	0.6
$L_{\text{mirror}}(\text{m})$	9.75
$W_{\text{mirror}}(\text{m})$	12.3
$H_{\text{tower}}(\text{m})$	130
$D_{\text{rec}}(\text{m})$	9.44
$L_{\text{rec}}(\text{m})$	9.44
α_{rec}	0.95
em_{rec}	0.85
$T_{\text{sun}}(\text{K})$	5770
HWU [43]	
T_{11}	25 °C
T_{12}	60 °C
MED [33]	
Number of effects	8
T_{20}	69 °C
T_{22}	25 °C
Feed water temperature (T_F)	35 °C
Seawater salinity	35000 ppm
Brine salinity	70000 ppm
SBC [43, 45]	
T_{16a}	35 °C
Compressor pressure ratio of SBC	2
Effectiveness of the SBC recuperator	0.86
$\eta_{isen,com}$	0.9

$\eta_{isen,GT}$	0.85
$T_9 - T_{10}$	100 °C
NH ₃ -LiNO ₃ ACH with ejector [35]	
$\eta_{isen,ACH,pump}$	0.95
Effectiveness of ACH SHX	.70
$T_{32} = T_{33}$	90 °C
T_{35}	35 °C
$T_{27} = T_{28}$	0 °C
T_{26}	35 °C
$T_{36} = T_{41}$	$T_{26} - 8$
$T_{37} = T_{42}$	$T_{26} - 3$
T_{38}	$T_{27} + 8$
T_{39}	$T_{27} + 3$
TEG [46]	
TEGs water inlet temperature	25 °C
TEGs water outlet temperature	35 °C
ZT_M	0.9
AE [47, 48]	
T_{AE}	60 °C
P_{AE}	7 bar
F	96,486 C mol ⁻¹
r_1	4.45153× 10 ⁻⁵
r_2	6.8884× 10 ⁻⁹
d_1	-3.12966× 10 ⁻⁶
d_2	4.47137× 10 ⁻⁷
s	0.33824
t_1	-0.01539
t_2	2.00181
t_3	15.24178

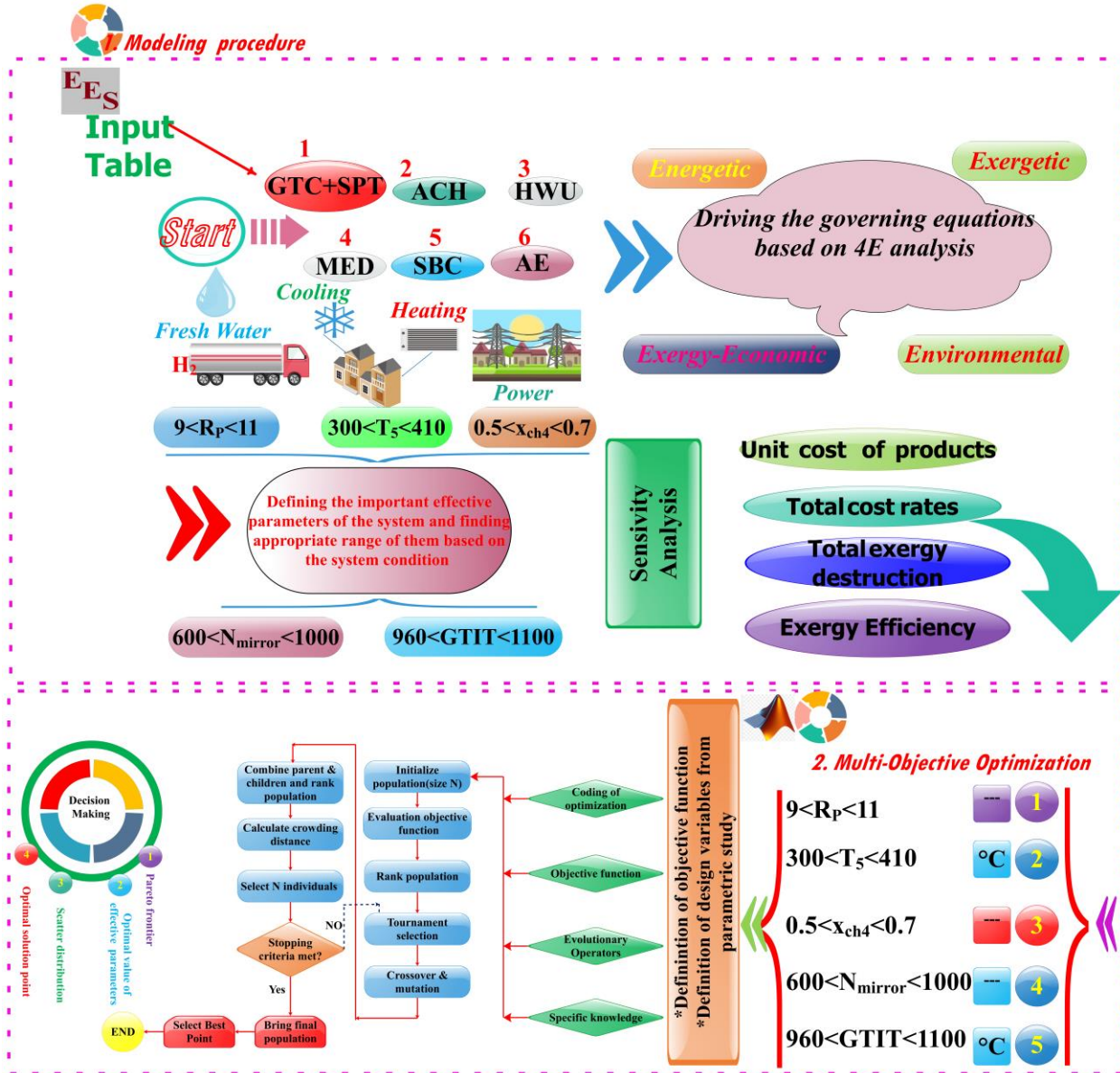


Fig. 6. The schematic diagram illustrating the procedural sequence for simulation and optimization processes

The evaluation of the energy, exergy, economic, and exergy-economic aspects of the system commences with the computation of the properties associated with each designated point. Table S3 in Supplementary Material delineates the thermo-economic parameters of each flow within the proposed system, including mass flow rate, temperature, pressure, enthalpy, entropy, exergy, and unit cost. These parameters were derived utilizing the EES software. Furthermore, the Sankey diagram illustrating the exergy flow within the system is presented in Fig. 7.

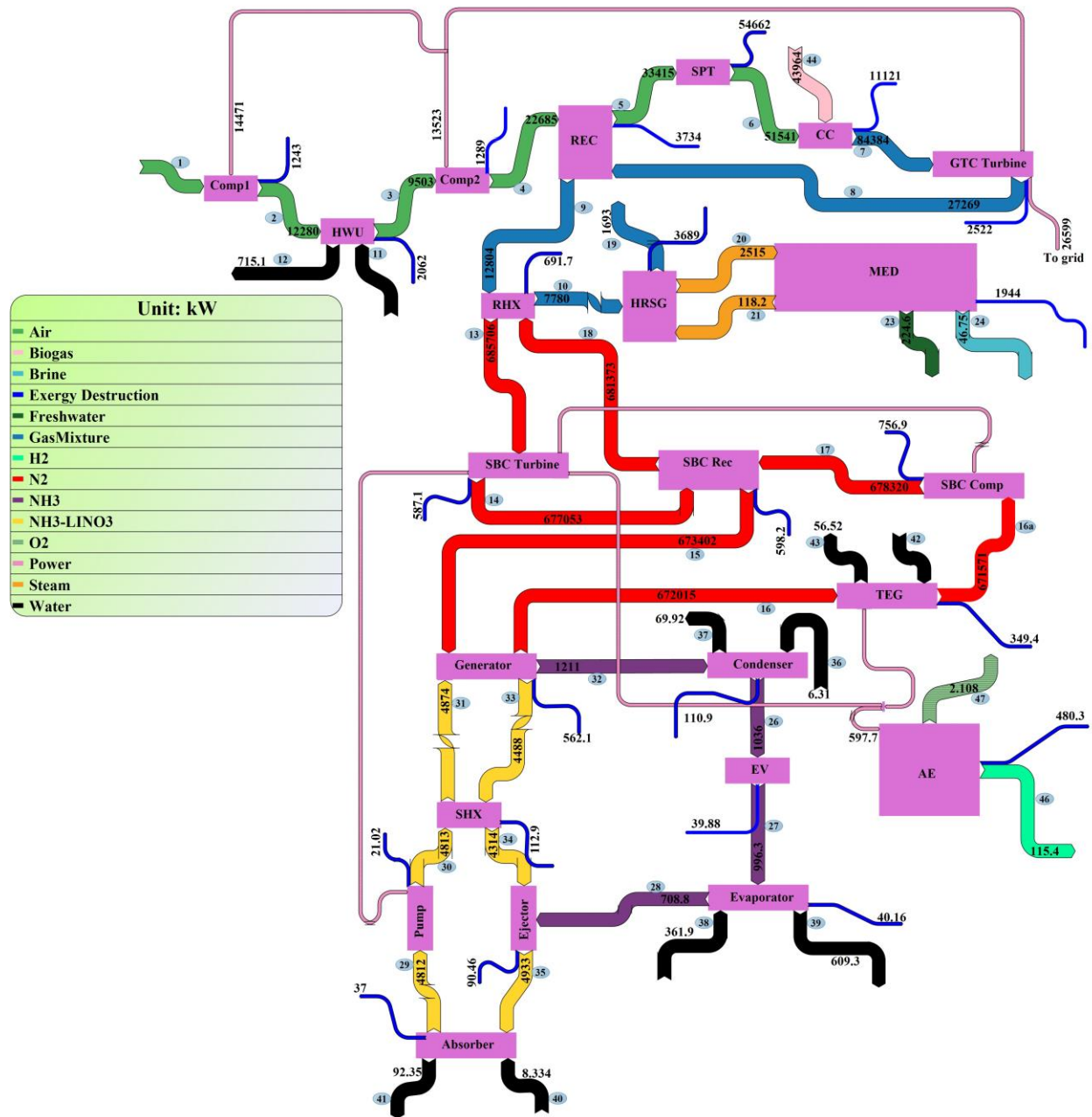


Fig. 7. The Sankey diagram of exergy flow

The output parameters that hold the most significant influence in the proposed poly-generation system at the base case are detailed in Table 6. It has been demonstrated that the GTC generates an output power of 26,599 kW, which constitutes the net output power, as indicated in Table 6. The recovery of waste heat from exhaust gas at 9 and 10 generates 3.474 kg/h of hydrogen in the AE and an astonishing 46.37 kg/s of fresh water in the MED. The recovery of waste energy from the compression stages and SBC, as well as the relevant procedures, resulted in 3550 kW of chilling in the ACH evaporator and 13015 kW of heating in the HWU. According to the total exergy rates of the fuel and product, the current system design has an exergy efficiency of 23.9%. It has been proved that among the various

cost rates, the exergy destruction cost rate, which is 4791 \$/h and accounts for about 48.68% of the total cost rate, yields the maximum value. Subsequently, the investment cost rate is calculated at 1029 \$/h, while the fuel cost rate is determined to be 3147 \$/h. The environmental cost rate, which has been determined to be 369.2 \$/h, is the value that is associated with the lowest numerical value. With a unit cost of 389.7 \$/GJ, freshwater is shown to have the highest value among the different unit prices. Consequently, it is established that the unit cost of hydrogen amounts to 191.7 \$/GJ. Significantly, the most cost-effective product is electricity, valued at 14.08 \$/GJ. Owing to the substantial share of electricity within the output products, the unit cost of poly-generation is calculated to be 19.42 \$/GJ.

Table 6 Primary performance metrics for the specified configuration

Parameters	Value
$\dot{W}_{GTC}(KW)$	26599
$\dot{W}_{net}(KW)$	26599
$\dot{W}_{SBC}(KW)$	597.7
$\dot{Q}_{heating}(KW)$	13015
$\dot{Q}_{cooling}(KW)$	3550
$\dot{m}_{freshwater}(\frac{kg}{s})$	46.77
$\dot{m}_{hydrogen}(\frac{kg}{h})$	3.474
$\eta_{ex,poly}(\%)$	23.9
$\dot{Z}_{tot}(\$/h)$	1029
$\dot{C}_{dest,tot}(\$/h)$	4791
$\dot{C}_{fuel}(\$/h)$	3147
$\dot{C}_{env}(\$/h)$	369.2
$\dot{C}_{tot}(\$/h)$	9337
$c_{el}(\$/GJ)$	14.08
$c_{heating}(\$/GJ)$	62.59
$c_{cooling}(\$/GJ)$	32.61
$c_{freshwater}(\$/GJ)$	389.7
$c_{hydrogen}(\$/GJ)$	191.7
$c_{poly}(\$/GJ)$	19.42

With respect to the Figs. 8-10, it is noteworthy to mention that GTC+SPT holds the largest share in $\dot{E}x_{dest,tot}$, $\dot{C}_{dest,tot}$ and $\dot{Z}_{tot}$ with respective contribution percentages of 82.65%, 64.2%, and 91.82%. Based on the data presented in Fig. 8, it can be inferred that there are two other crucial subsystems, namely SBC and ACH, which altogether account for a significant proportion of the total exergy that has been destructed. Following the GTC+SPT,

it can be observed that the SBC holds the highest potential for contributing to the overall exergy destruction, accounting for a significant proportion of 7.4% of the total. The subsequent systems that follow closely in the contribution include the ACH and the HWU systems, which represent 4.98% and 2.29% of the total exergy destruction, respectively. Therefore, it is apparent that these subsystems warrant thorough examination and should be accorded significant attention when assessing the comprehensive performance and operational efficacy of the system in question. With reference to Fig.9, it is evident that the second bold system aligns with the ACH, accounting for 18.19% of the total exergy destruction cost rate. It is significant to highlight that the remaining systems demonstrate contribution percentages that do not exceed 10% in the aggregate exergy destruction cost rate. Based on the findings presented in Fig. 10, it can be concluded that subsequent to GTC+SPT, the SBC exhibits the most significant contribution percentage in the investment cost rate, amounting to 7.21%.

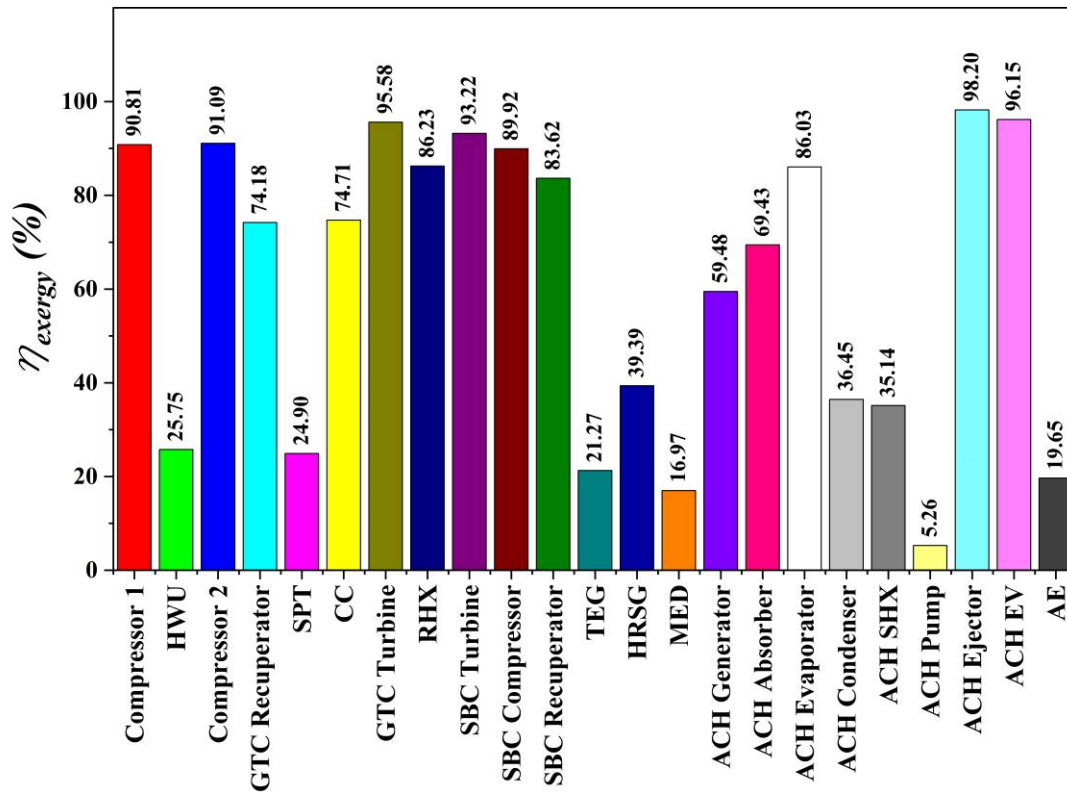


Fig. 8. The various components' exergy efficiency

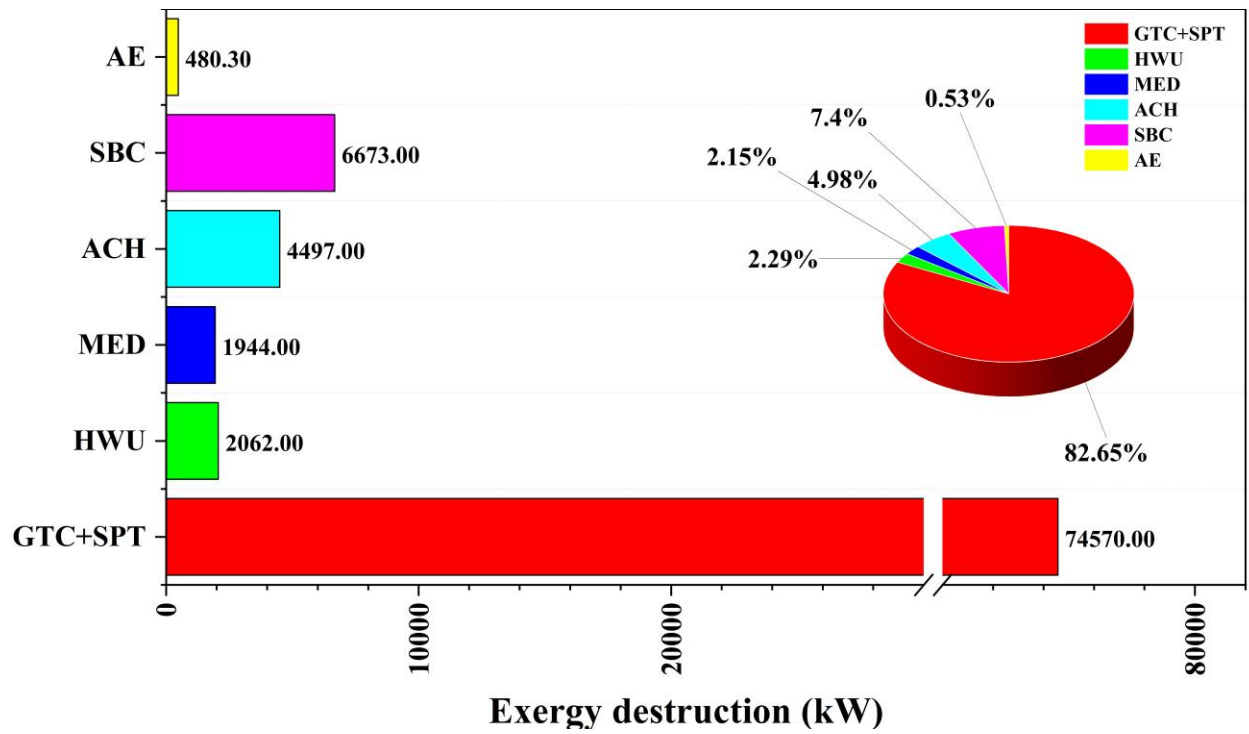


Fig. 9. The importance and proportion of exergy destruction in various subsystems

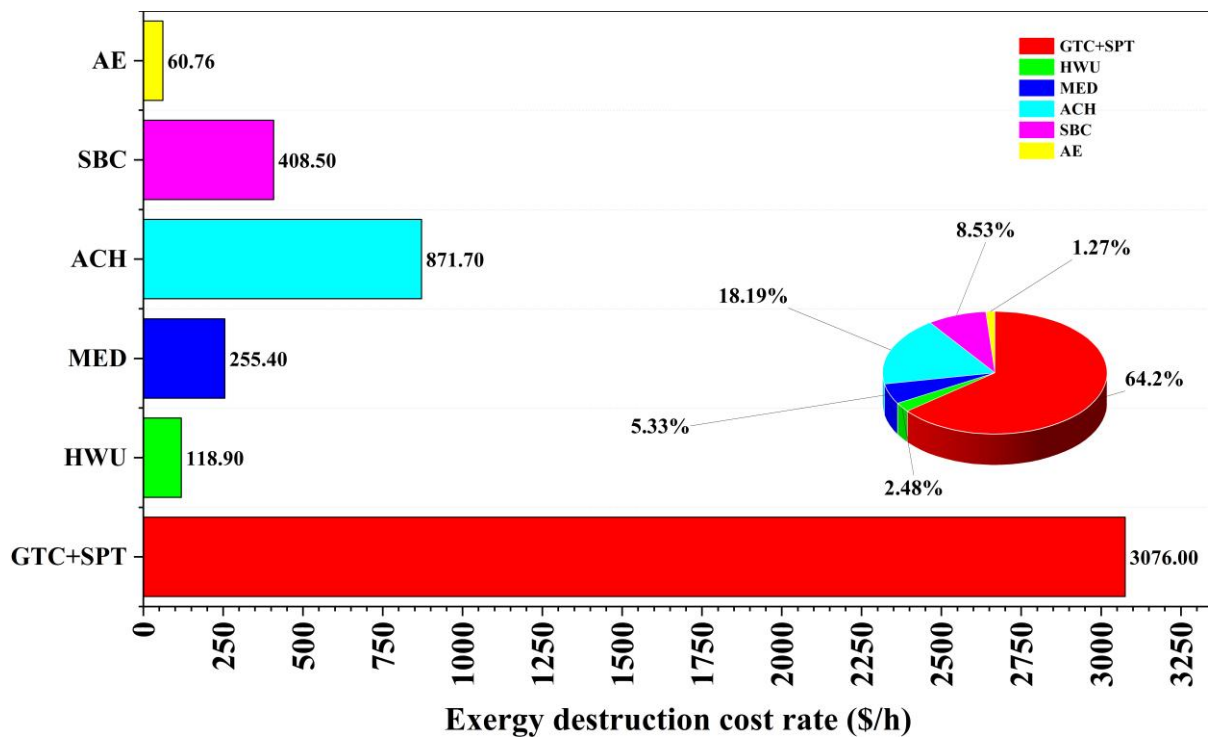


Fig. 10. The exergy destruction cost rate's value and proportion in various subsystems

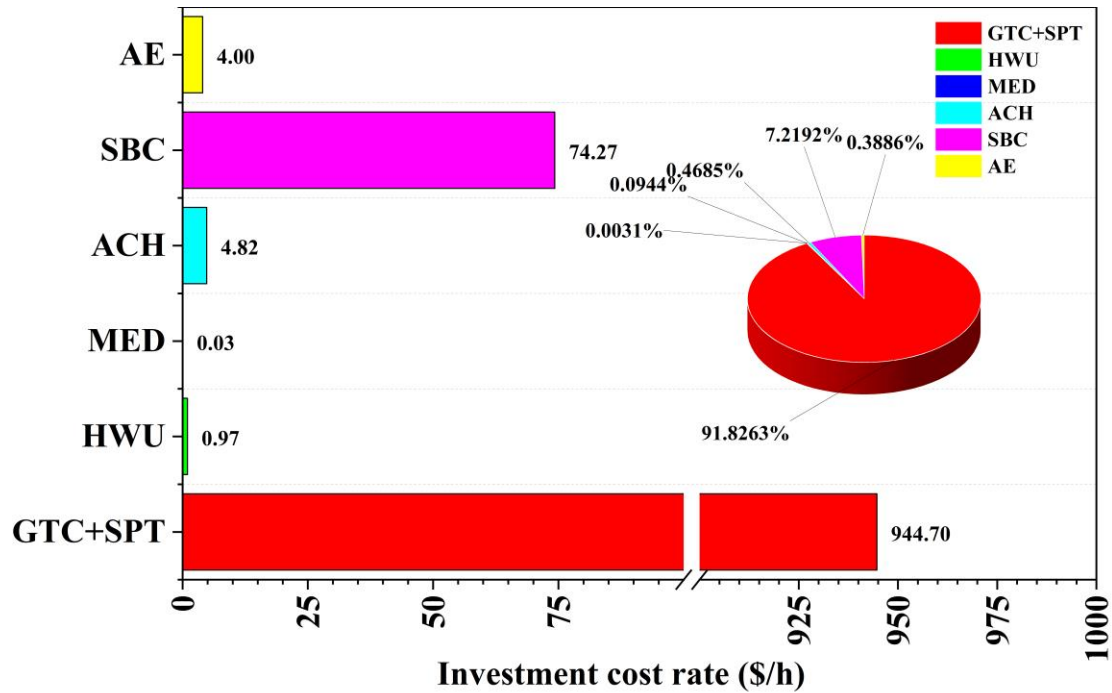


Fig. 11. Cost analysis and distribution metrics of investment expenditures in subsystem frameworks

4.2. Parametric Analysis

This section evaluates various performance metrics, such as power generation, thermal energy output, cooling capacity, freshwater and hydrogen production rates, $\eta_{ex,poly}$, $\dot{C}_{tot}$, and c_{poly} , in response to changes in the pressure ratio (R_p) of the GTC, mirror quantity, methane concentration in biogas, gas turbine inlet temperature (GTIT), and the input temperature of the SPT (T_5). During parametric analysis, one parameter is varied while others remain constant, as shown in Table 5.

Fig. 12 demonstrates the effect of GTC's R_p (4 to 11) on system outputs. Increasing R_p raises the temperature at point 2 and pressure at point 7, enhancing the HWU's heating output and GTC's net power. Cooling capacity improves by 3.7% due to a slight increase in refrigerant mass flow (state point 27) relative to R_p , as shown in Fig. 12a. However, freshwater production decreases as higher R_p reduces T_9 , lowering motive steam mass flow in the MED process (Fig. 12b). Hydrogen production drops by 55.7% due to reduced SBC net power. Exergy efficiency improves (16.19% to 24.39%, Fig. 12c) with increased power generation, heating exergy, and product exergy rates. The system's investment cost rate rises from 958.4 \$/h to 1038 \$/h (Fig. 12d) due to higher GTC and SBC turbine costs. Exergy destruction in the MED system decreases, while its cost rate in the absorber and SBC compressor increases, minimizing total exergy destruction cost at $R_p \approx 11$ (Fig. 12e). The total cost rate reaches a minimum at $R_p \approx 10.3$, influenced by investment costs and exergy destruction trends. The c_{poly} decreases from 27.95 \$/GJ to 19.05 \$/GJ (Fig. 12f), driven by lower freshwater and hydrogen production costs.

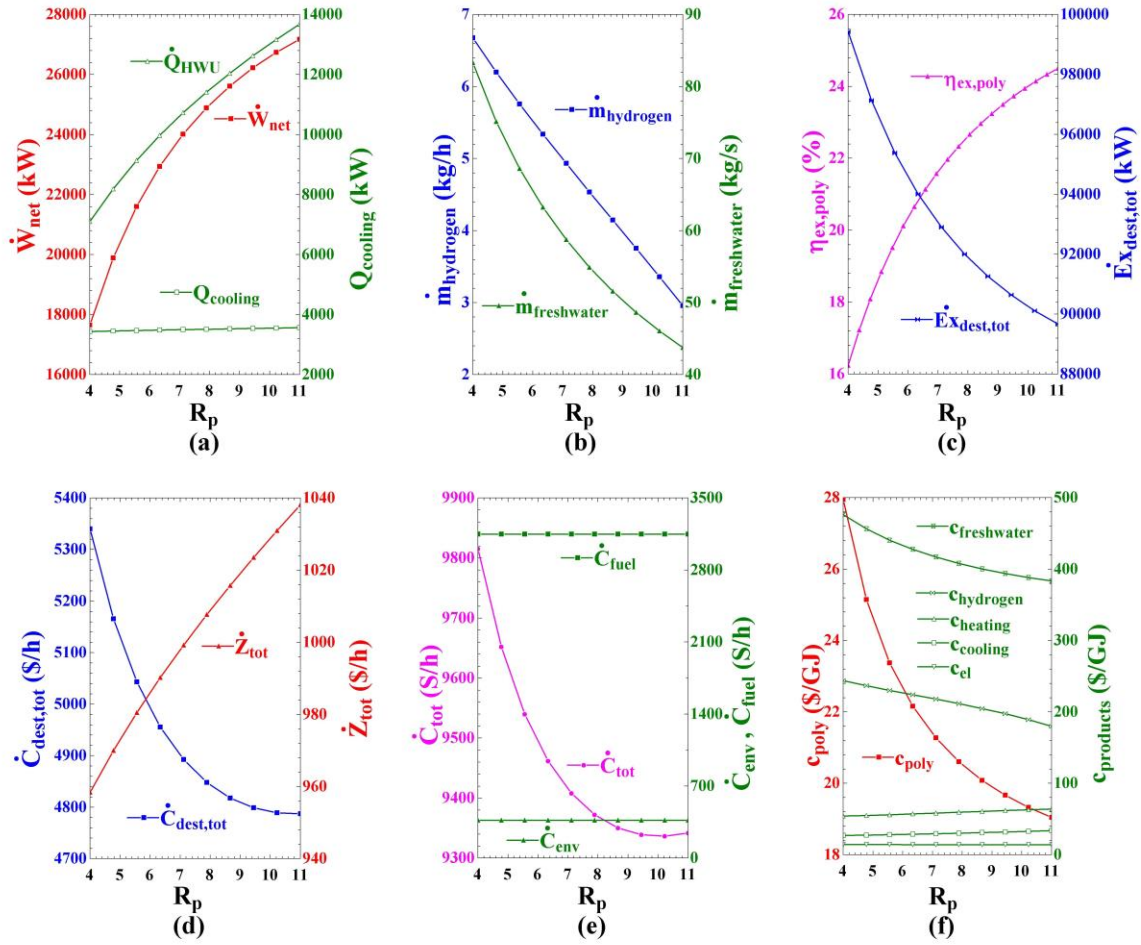


Fig. 12. The impact of GTC's R_p on output variables

The effect of increasing mirror quantity from 600 to 1000 on system performance is shown in Fig. 13. Higher mirror numbers elevate combustion chamber air temperature, reducing the enthalpy difference across the gas turbine and decreasing its mass flow rate. This lowers the GTC net output power by 2.46% (Fig. 13a), alongside 5.5% and 9.19% declines in hydrogen and fresh water production, respectively. Reduced electrolyzer power consumption further decreases hydrogen mass flow rate (Fig. 13b). Increased mirrors also cause a 16.4 percentage point drop in overall exergy efficiency due to lower exergy production (Fig. 13c). Key concerns include reduced net power output, fresh water, and hydrogen generation. Mirror augmentation raises total exergy destruction by 23.8%, mainly from SPT, while exergy cost rates in HWR, AE, and the combustion chamber decrease by 4.5% (Fig. 13d). Expanding mirror area increases SPT capital expenditure, driving higher system investment costs. Reduced fuel consumption and CO₂ emissions lower overall costs (Fig. 13e), and decreased hydrogen and water unit costs reduce poly-generation costs by 11.62% (Fig. 13f).

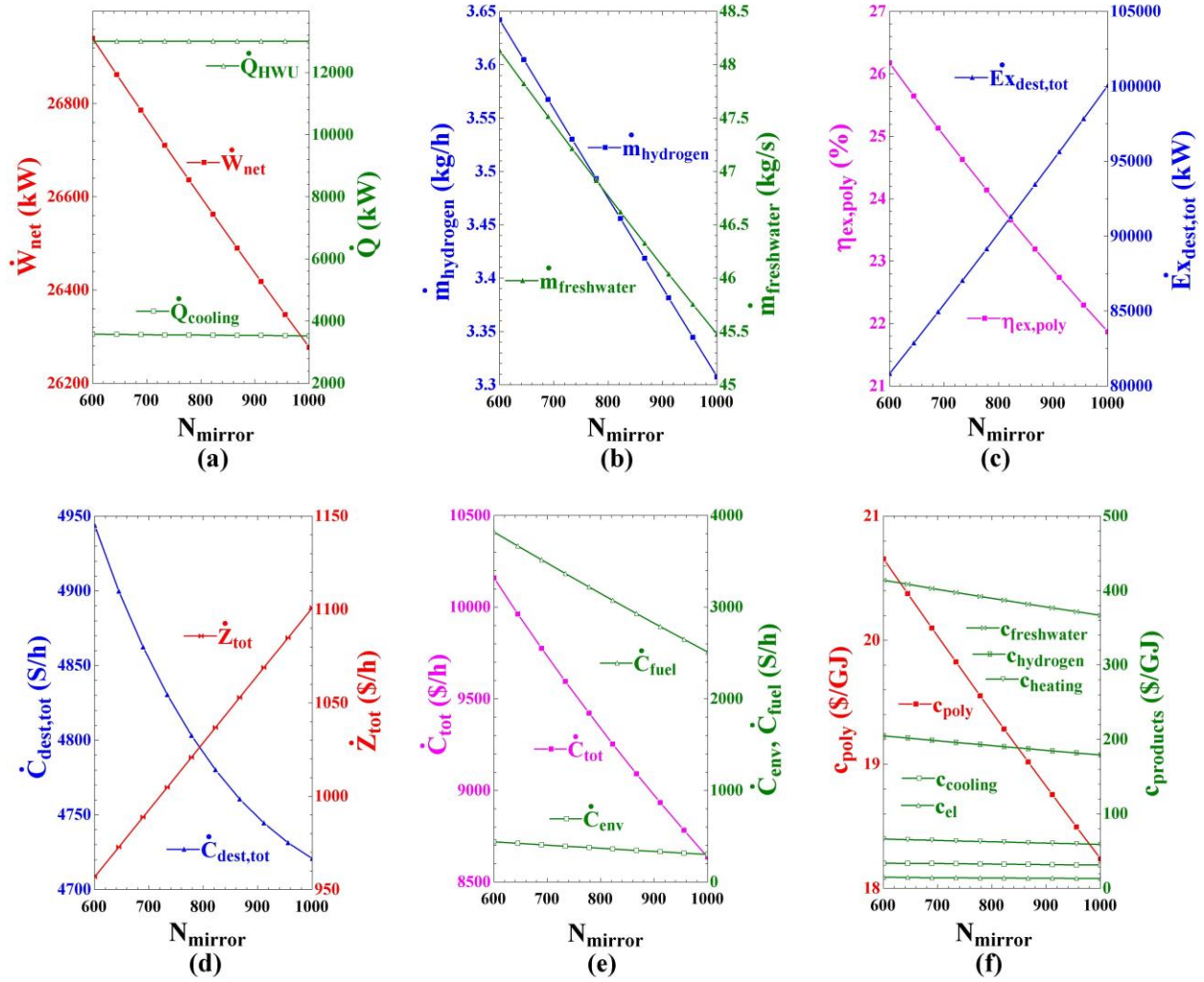


Fig. 13. The impact of mirror count on the output parameters

Variations in GTIT significantly impact system performance, as shown in Fig. 14. The net output power of the GTC subsystem increases with higher GTIT due to a greater enthalpy difference at the turbine's inlet and outlet. However, thermal energy production and GTC compressor power consumption remain unchanged. Increasing GTIT raises the RHX inlet temperature in the GTC subsystem while keeping the ACH generator temperature at 90 °C. The cooling capacity slightly decreases by 1.6% due to a reduced SBC mass flow rate (Fig. 14a). Freshwater and hydrogen production, driven by the motive steam flow rate ($\dot{m}_{21}$) in the MED system and higher SBC net output power, positively correlate with GTIT (Fig. 14b). The net output power of the SBC and AE power consumption are interdependent. Overall $\eta_{\text{ex,poly}}$ shows a significant upward trend (Fig. 14c), primarily due to increased net power output and hydrogen production exergy. $\dot{E}x_{\text{dest,tot}}$ rises from 86,362 kW to 100,381 kW, mainly due to the combustion chamber's efficiency and MED system exergy destruction. Higher GTIT also increases investment costs for the combustion chamber, AE, and HRSG while negatively affecting the SBC compressor and turbine performance. The investment cost rate reaches its lowest point at GTIT = 1050 °C (Fig. 14d). The overall exergy degradation cost rate

exhibits an upward trend (Fig. 14d), driven by exergy destruction in the combustion chamber, HRSG, and MED unit. The total cost rate increases from 8830 \$/h to 10882 \$/h due to rising exergy destruction costs in fuel and environmental factors (Fig. 14e). Additionally, poly-generation unit cost rises by 4.7% when hydrogen and freshwater unit costs exceed certain thresholds (Fig. 14f).

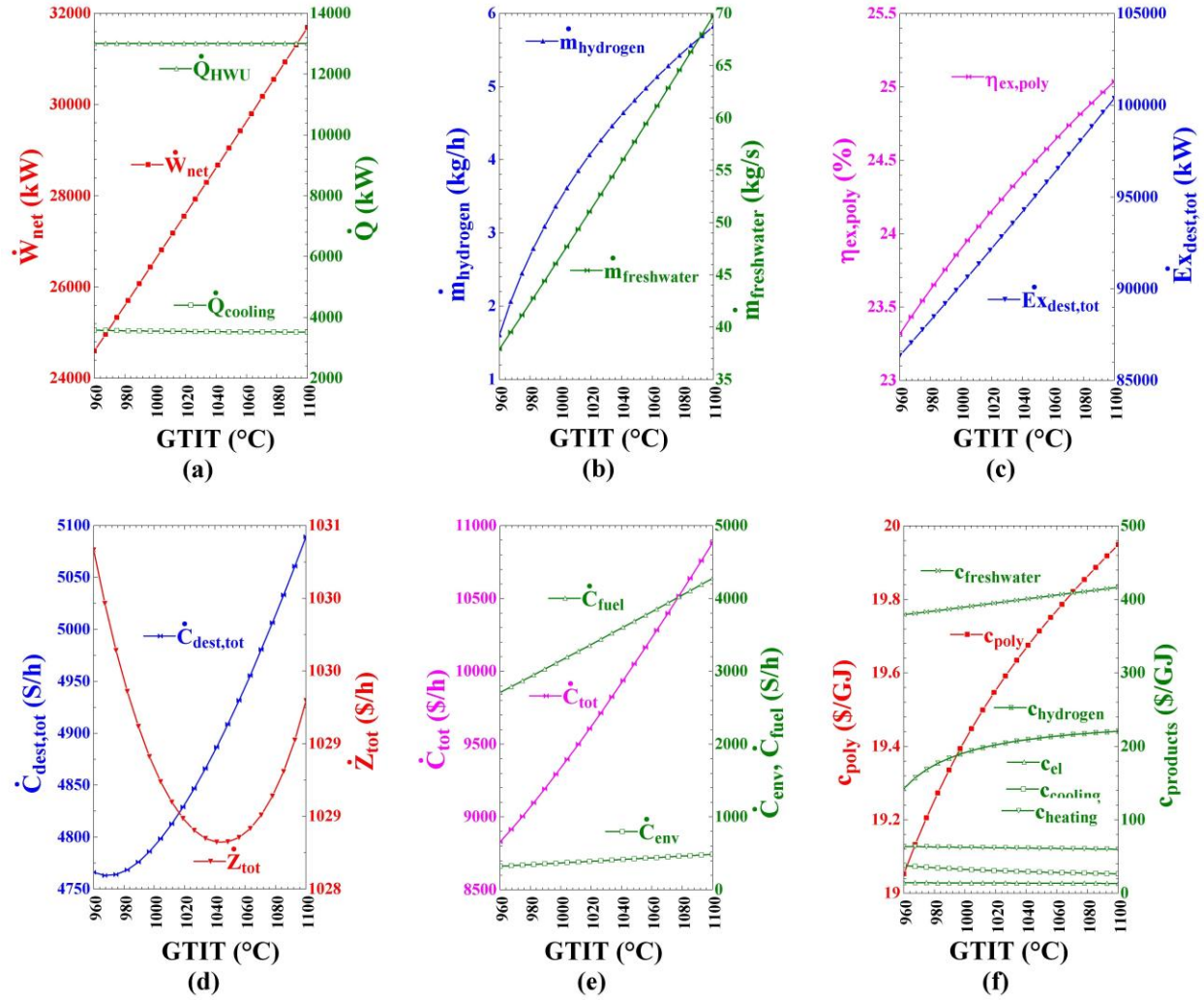


Fig. 14. The effect of GTIT on the output parameters

The impact of methane molar fraction (x_{ch_4}) variations, ranging from 0.5 to 0.7, on system performance is depicted in Fig. 15. As shown in Fig. 15a, an increase in x_{ch_4} reduces the gas mixture's mass flow rate and GTC turbine power output, lowering net power. Fig. 15b indicates a drop in fresh water production from 48.02 kg/s to 45.91 kg/s as x_{ch_4} rises, due to decreased MED steam flow. Additionally, the RHX gas inlet temperature (T_9) declines with higher x_{ch_4} reducing the SBC turbine's enthalpy differential and net power output. This decrease lowers electrolyzer power consumption, resulting in a 7.17% reduction in hydrogen production. Overall exergy efficiency slightly declines (Fig. 15c), driven by reduced net power and hydrogen synthesis exergy, decreasing multi-generation efficiency by 0.2

percentage points. $\dot{E}x_{dest,tot}$ also falls from 91033 kW to 89675 kW. Fig. 15d shows that x_{ch_4} variations affect exergy destruction and investment cost rates, with both declining alongside MED exergy destruction. The overall cost rate decreases by 19.4% due to lower investment, exergy destruction, and fuel exergy costs (Fig. 15e). Fig. 15f reveals a minimal reduction in the system's unit cost as x_{ch_4} increases.

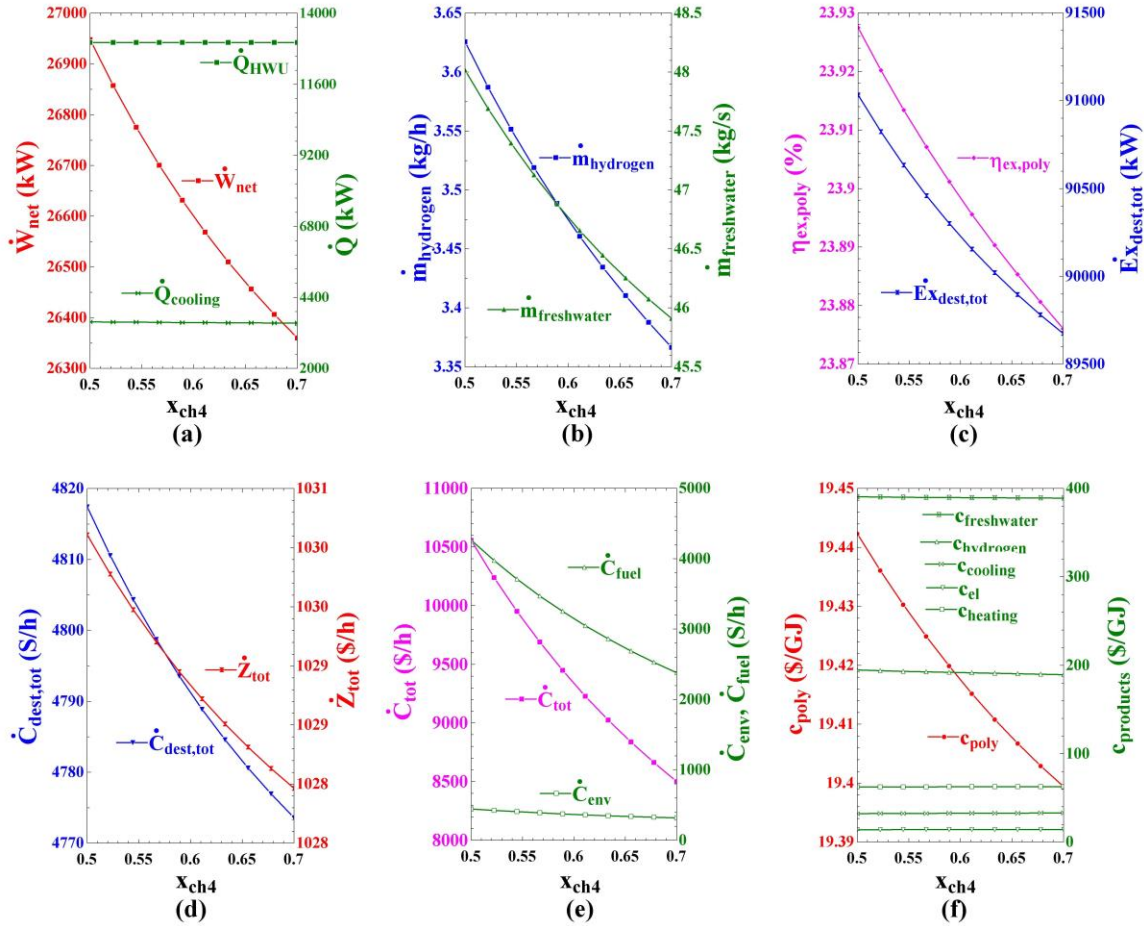


Fig. 15. Methane molar fraction's impact on the output parameters

Fig. 16 examines the effect of elevating the solar power tower (SPT) inlet temperature (T_5) from 300°C to 410°C on thermodynamic and exergy-economic performance. As T_5 rises, the gas turbine cycle (GTC) net output power decreases from 26,934 kW to 26,566 kW due to a reduced gas turbine's enthalpy differential, as shown in Fig. 16a. Conversely, increasing the SBC mass flow rate slightly enhances cooling production. Fig. 16b reveals a significant drop in freshwater production from 74.56 kg/s to 43.96 kg/s, attributed to reduced MED motive steam flow caused by higher SPT inlet temperatures. Additionally, the SBC's net power demand for electrolyzer operation decreases substantially, cutting hydrogen production mass flow by 51.1%. Exergy efficiency rises by 6.37% (from 22.59% to 24.03%, Fig. 16c) due to lower fuel exergy flow rates, while total exergy destruction declines. Fig. 16d shows increased investment costs for the GTC recuperator, compressor, and turbine, alongside rising exergy destruction costs

for the TEG, SBC compressor, and ACH heat exchanger. However, MED and combustion chamber exergy destruction costs decrease, with total destruction costs ($\dot{C}_{dest,tot}$) reaching a minimum near $T_5 = 398^\circ\text{C}$. Fig. 16e highlights a reduction in the overall cost rate from 10,253 \$/h to 9,274 \$/h due to lower fuel consumption costs. Fig. 16f indicates declining unit production costs for electricity, heating, cooling, hydrogen, and freshwater, driven primarily by reduced freshwater and hydrogen production costs.

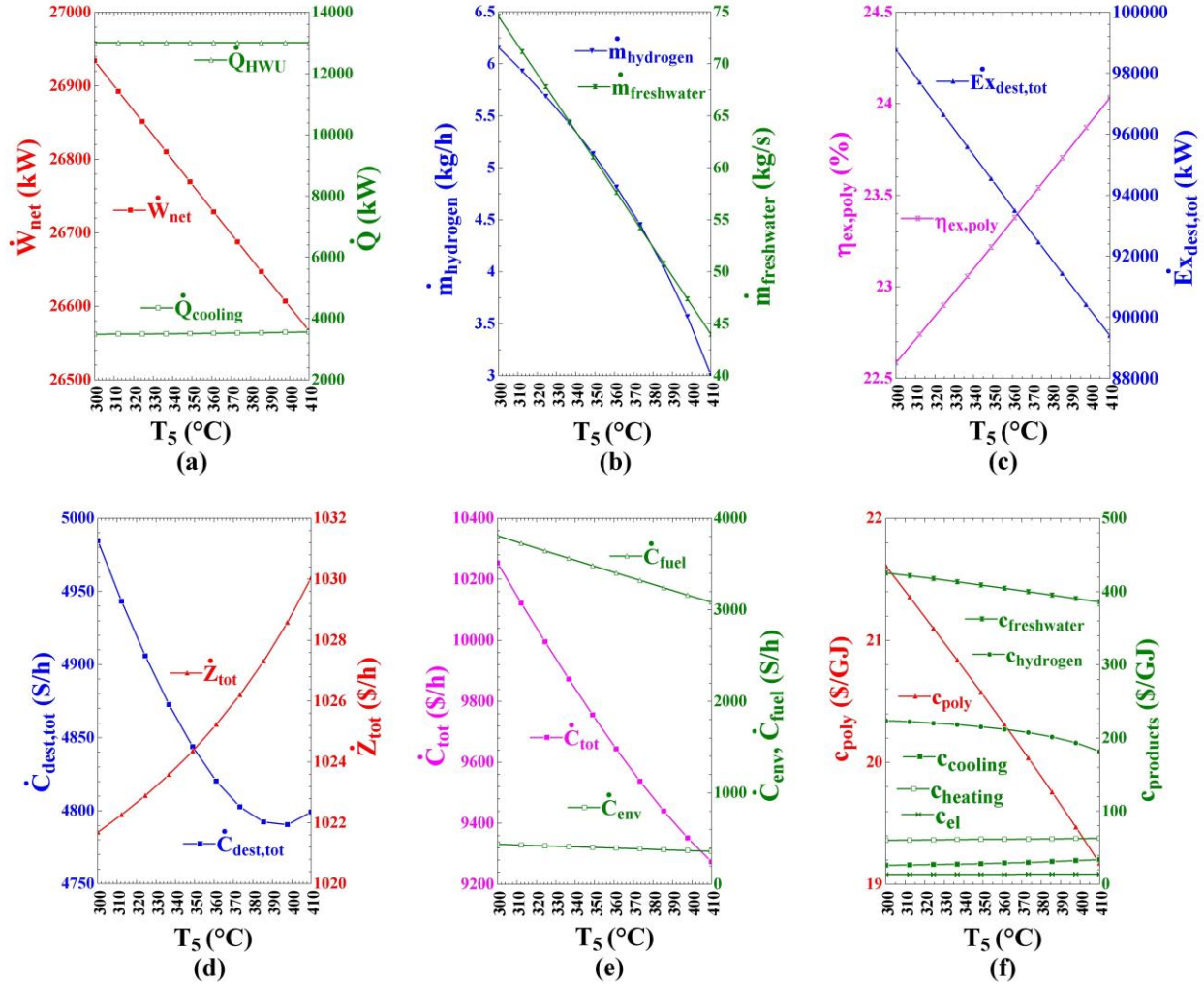


Fig. 16. SPT intake temperature's impact on output parameters

4.3. The findings derived from the optimization analysis

This section delineates the results obtained from the NSGA-II algorithm, accompanied by the optimization process involving three distinct objectives. The objectives of $\eta_{ex,poly}$, $\dot{C}_{tot}$, and c_{poly} , which are in conflict with each other, have been taken into account. The Pareto optimal frontier and corresponding numerical results are presented in Fig. 17 and Table 7. The blue circles positioned along the periphery represent the optimal locations that remain unobstructed by overlapping data points. Following this, Table 7 delineates the results corresponding to the trade-off

point (indicated by the green point), which are subsequently compared with the outcomes derived from the baseline scenario. As per the calculations, at the optimal point, the values of $\eta_{ex,poly}$, $\dot{C}_{tot}$, and c_{poly} are determined to be 25.10%, 8482.67 \$/h and 19.01 \$/GJ, respectively. Upon comparison with the base case, enhancements of 5.02%, 9.14%, and 2.11% have been documented for $\eta_{ex,poly}$, $\dot{C}_{tot}$, and c_{poly} , respectively. Additionally, Fig. 18 illustrates the scatter distribution of five selected parameters in relation to population size, as determined by the NSGA-II optimization method. It is significant to note that the graphical representation offers critical insights into the dynamics of the decision variables as the population size varies. Consequently, this visualization serves as a valuable tool for enhancing the understanding of the optimization process and facilitating well-informed decision-making. For further elaboration, error histogram diagrams are presented in Fig. 19.

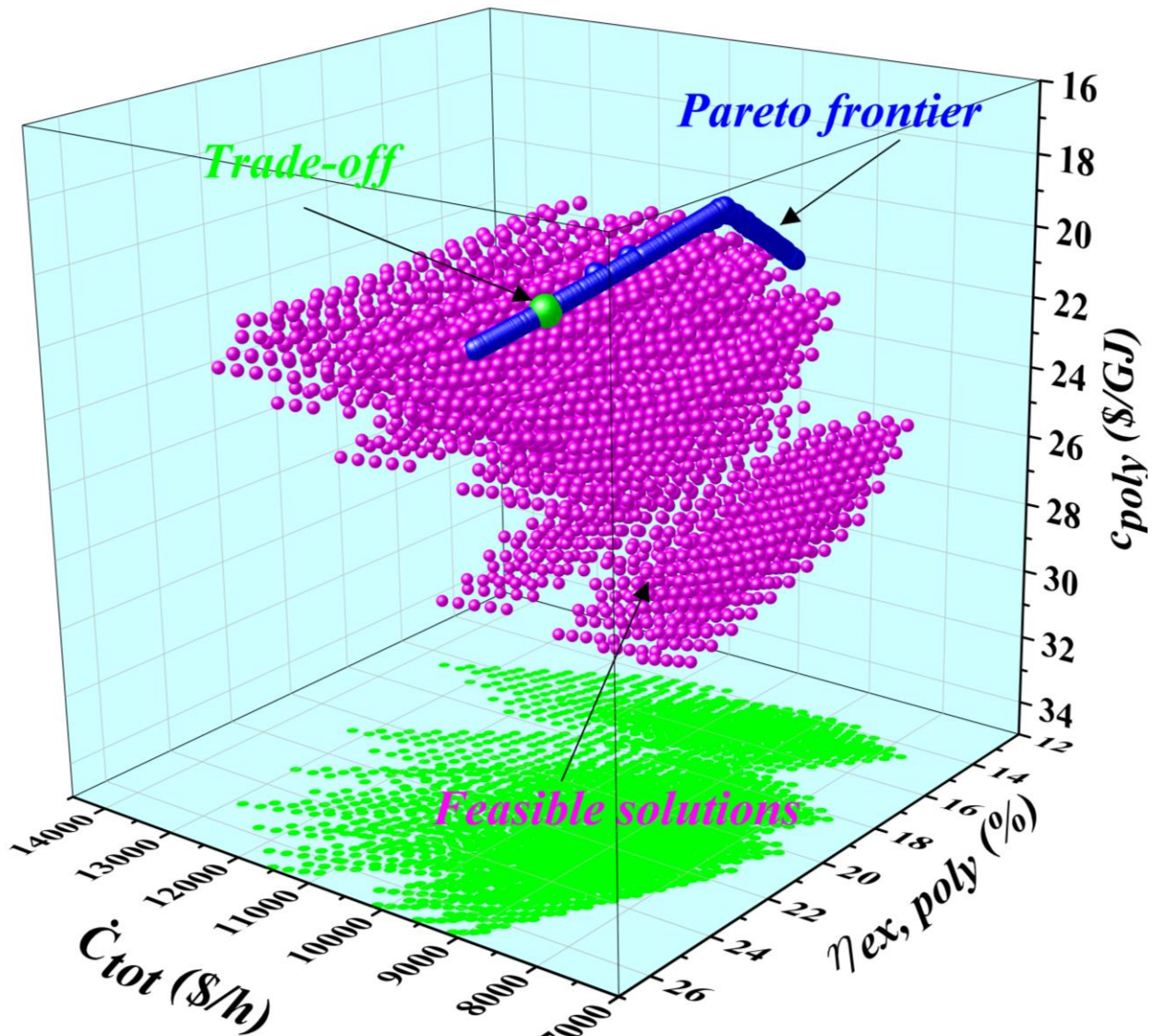
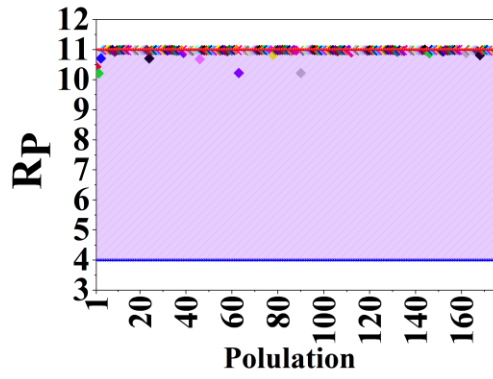


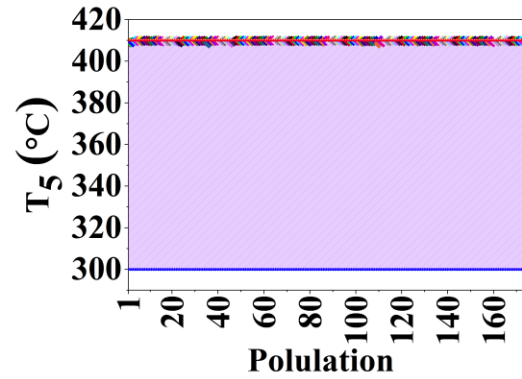
Fig. 17. Generated pareto optimal frontier visualization

Table 7 Findings from the optimization analysis

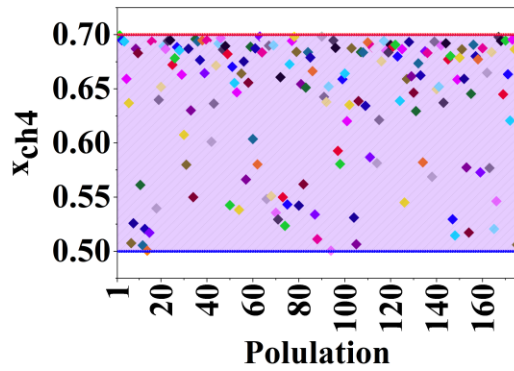
Parameters (Design variables and objective functions)	Base case	Optimal value (trade-off)	Performance improvement (%)
R_p	10	11	–
$T_5(^{\circ}\text{C})$	400	410	–
x_{ch_4}	0.6	0.7	–
$GTIT(^{\circ}\text{C})$	1000	960	–
N_{mirror}	800	696	-
$\eta_{ex,poly}$ (%)	23.9	25.10	5.02%
$\dot{C}_{tot}$ (\$/h)	9337	8482.67	9.14%
$c_{poly}(\$/GJ)$	19.42	19.01	2.11%



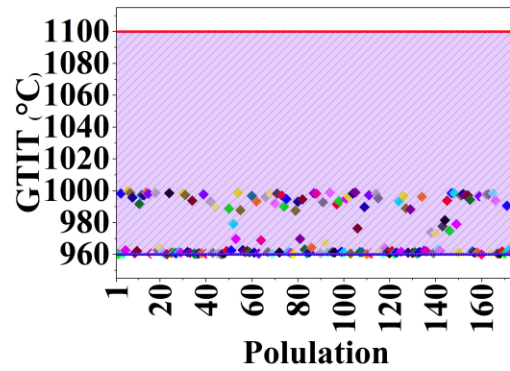
(a)



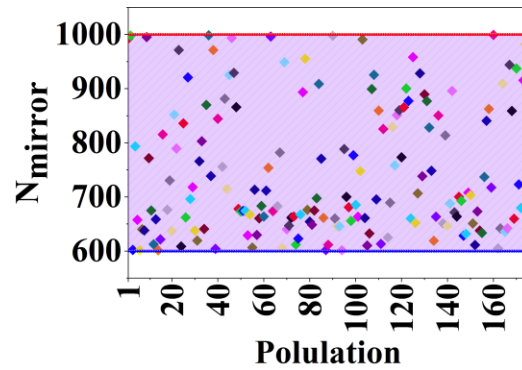
(b)



(c)



(d)



(e)

Fig. 18. Scatter distribution for the design variables

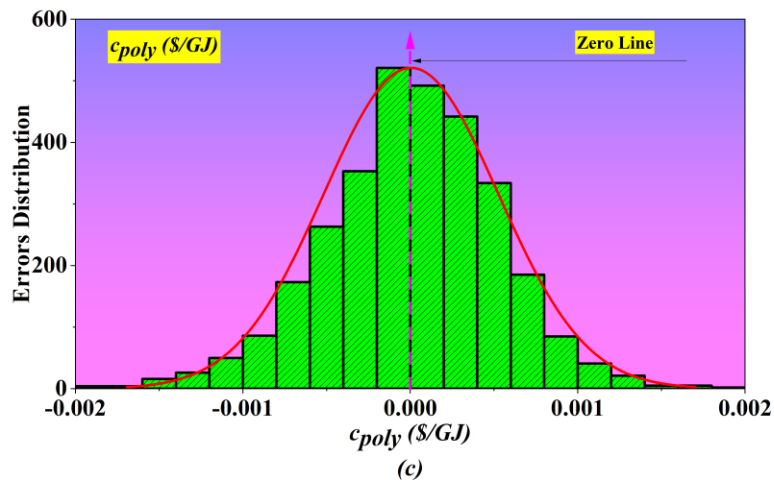
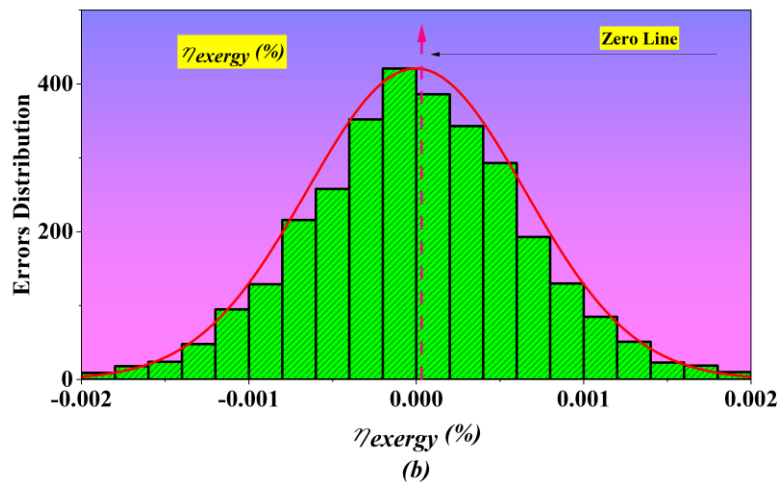
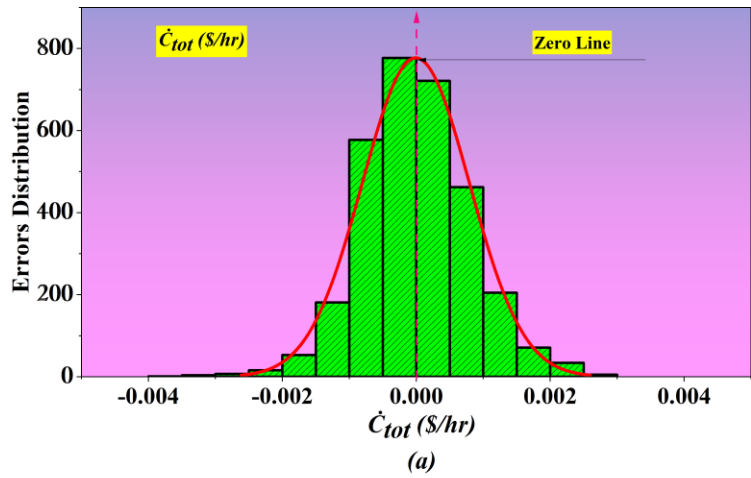


Fig. 19. ANN model histogram plot errors

The investigation focused on an ANN configured with a range of neuron counts. The Mean Squared Error (MSE) was computed for each newly constructed network, with particular emphasis placed on the average MSE associated with different neuron quantities. Fig. 20 illustrates the relationship between MSE values and the number of neurons, revealing that an optimal configuration was achieved with 25 neurons. Fig. 21 presents a comparative analysis between the actual data, derived from normalized outputs, and the predicted data generated by the ANN. The methodology employed demonstrates a high degree of consistency and a strong correlation with numerical results. The residual error reaches its minimum value at approximately -0.002. Fig. 22 displays the scatter distribution matrix, indicating that the temperature T_5 is around 410 degrees Celsius, as depicted in the image.

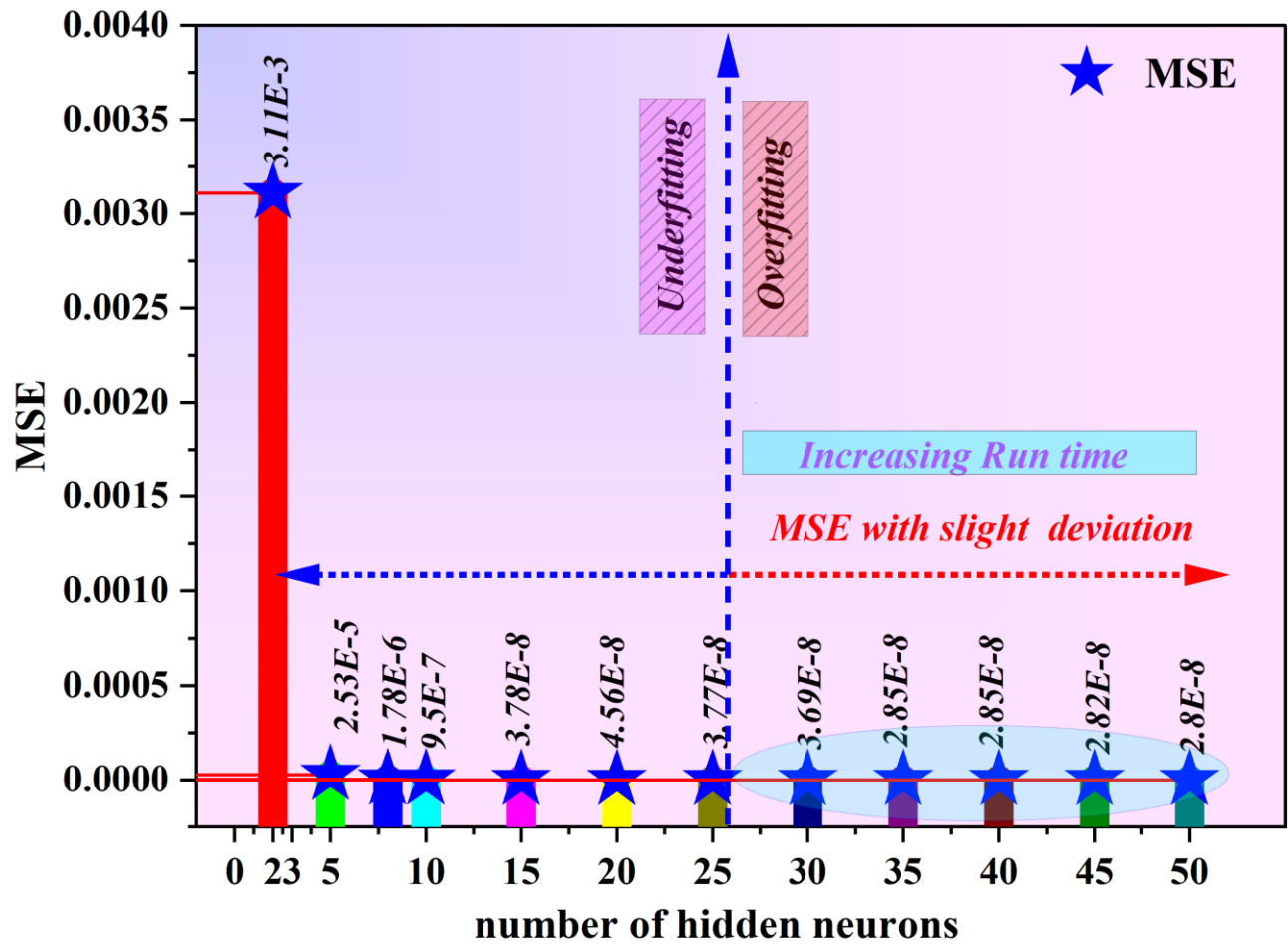


Fig. 20. MSE and hidden neuron configuration in Diverse Artificial Neural Network networks

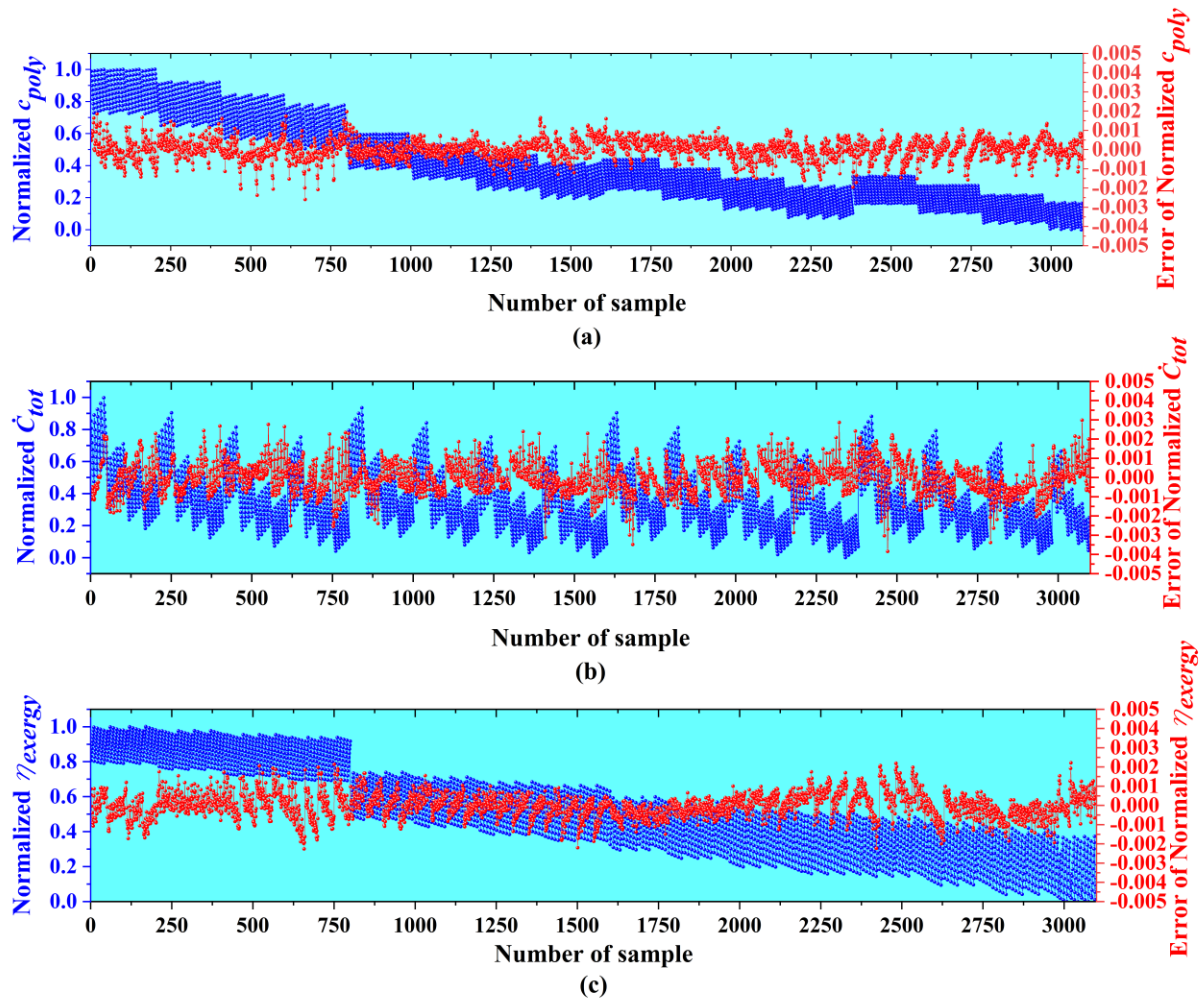


Fig. 21. Errors estimated in the proposed ANN system model

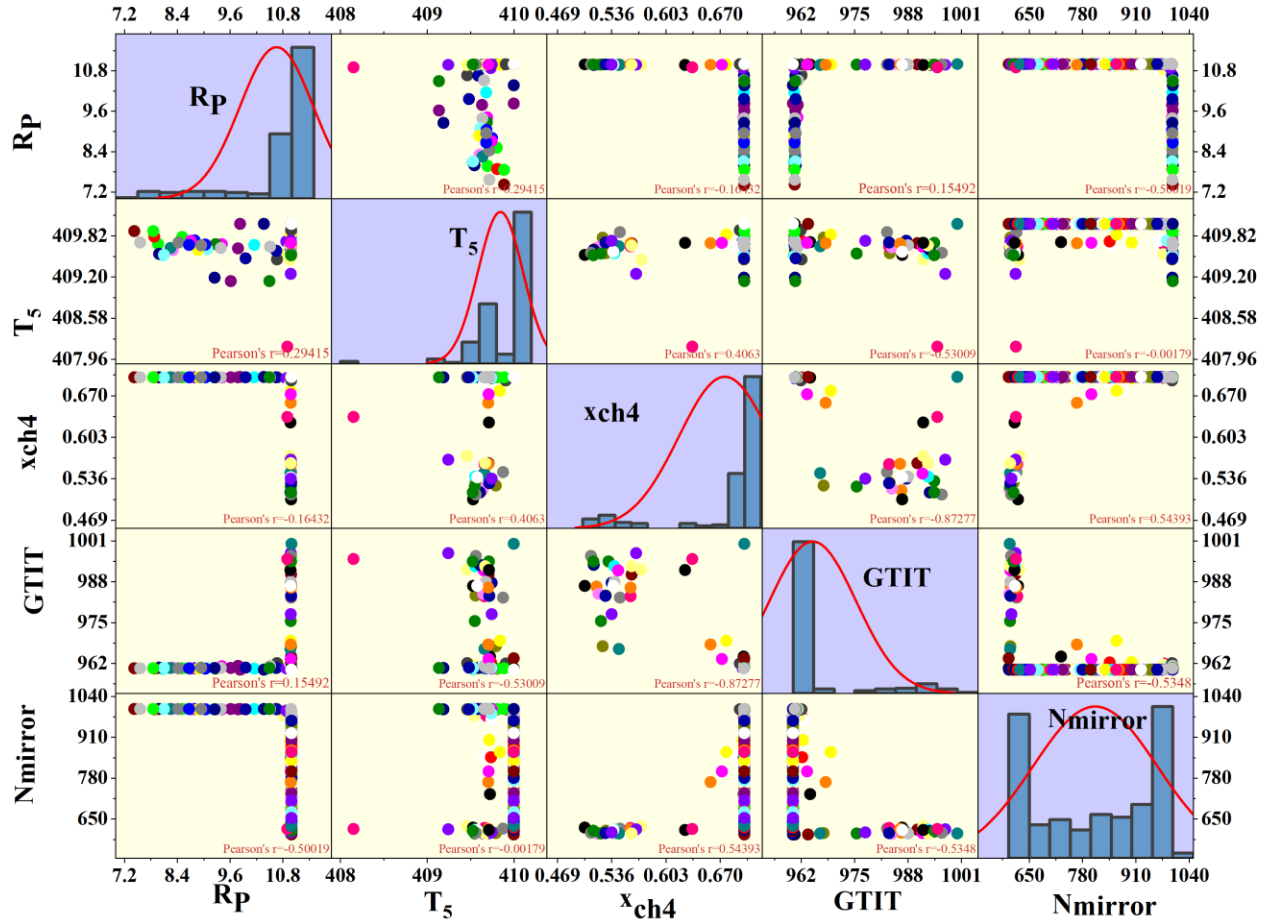


Fig. 22. The selection criteria for this study's scatter distribution

Moreover, Fig. 23 illustrates the pairwise and normalized correlations among the performance metrics and the input parameters utilized in this evaluation. A linear relationship between the two variables under investigation is indicated by values closer to 1 or -1, while the sign of the correlation coefficient denotes the direction of the relationship. Specifically, $\dot{C}_{tot}$ and R_p exhibit a negative correlation (-0.14), whereas η_{ex} and T_5 demonstrate a positive correlation (0.15).

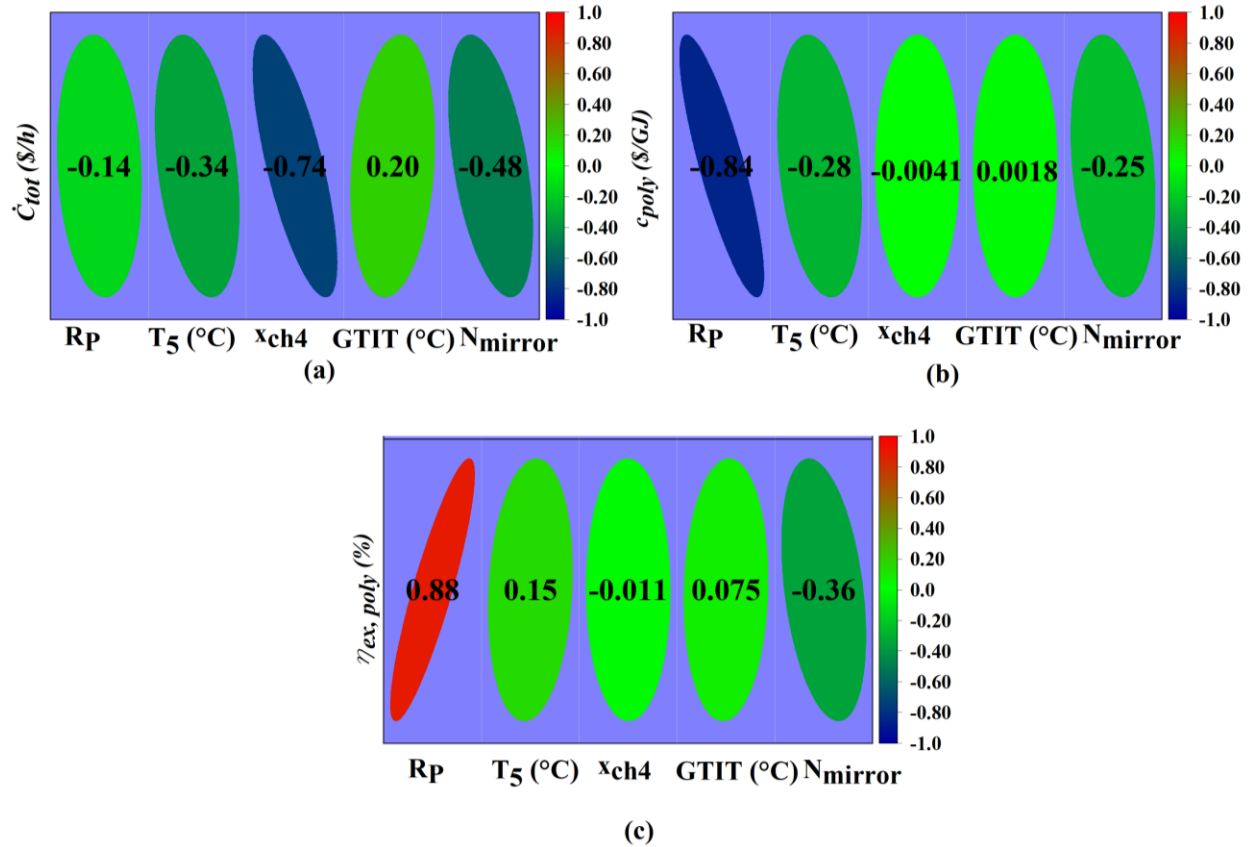


Fig. 23. The performance requirement and inputs' pairwise connection, as well as the correlation matrix

5. Comparative examination

A comparative examination of the output outcomes from pertinent research on solar-hybrid multi-product configurations and the results obtained in the current investigation is delineated in Table 8. As evidenced by the findings, the outcomes of the current investigation exhibit performance that is competitive with prior research, especially in the context of hybrid SPT and biogas combustion chamber heat source configurations. Significantly, within the realm of multi-generation systems designed to generate power, heating, cooling, freshwater, and hydrogen (as referenced in studies [49] and [50]), the present study achieves higher exergy efficiency and a reduced unit cost in comparison to these closely aligned studies.

Table 8 Multi-product systems based on solar hybrid heat sources

Reference	Heat source	Topping system	Bottoming systems	Products	Exergy efficiency (%)	Unit cost of outputs (\$/GJ)
[7]	SPT + Methane CC	GTC	Air bottoming cycle + HRSG + ORC + ACH	Power + Heating + Cooling	24.74	18.63
[8]	Biomass gasifier CC + SPT	GTC	Cascade SRC/ORC + MED + PEM electrolyzer	Power + Freshwater + Hydrogen	26.7	35.7

[9]	SPT + Methane CC	GTC	SRC/Flash desalination + ORC/TEG 1 + ACH/TEG 2	Power + Cooling + Fresh water	42.8	–
[10]	SPT + Biomass gasifier CC	GTC	CO ₂ SBC + TEG + PEM electrolyzer + ACH + HRSG	Power + Heating + Cooling + Hydrogen	43.12	8.34
[51]	SPT + Biomass gasifier CC	GTC	SRC + TEG + PEM electrolyzer + HWU + ACH	Power + Heating + Cooling + Hydrogen	43.11	8.26
[49]	SPT + Biogas CC	GTC	Geothermal flash cycle + Power and cooling ejector cycle + HWU + 2 TEGs + AE + RO desalination	Power + Heating + Cooling + Freshwater + Hydrogen	18.36	32.2
[50]	SPT + Biogas CC	GTC	Cascade SRC/ACH + Organic flash cycle + HWU + PEM electrolyzer + RO desalination	Power + Heating + Cooling + Freshwater + Hydrogen	24.34	20.17
Present study	SPT + Biogas CC	GTC	Nitrogen SBC + TEG + Ejector-based ACH + HWU + AE + MED	Power + Heating + Cooling + Freshwater + Hydrogen	25.1	19.01

6. Conclusion

In the current investigation, a novel poly-generation arrangement was introduced, which relies on the combination of a GTC and an SBC. The GTC was fueled by solar and biogas energies, with the aim of reducing the limitations of each energy source. Additionally, a HWU was employed to recover wasted energy from GTC compressor stages, whereas a MED unit and an SBC were utilized to eliminate waste heat from GTC and produce freshwater and cooling, respectively. To cater to the basic needs of human beings, the SBC's net output power was conveyed to an AE to produce hydrogen. A thorough 4E study, comprising parametric analysis, design optimization, and a base case, was performed on the system. The following summarizes the main findings from the study:

In the base-case scenario, the proposed design may produce 26599 kW of electricity, 13015 kW of heating, 3550 kW of cooling, 46.77 kg/s of fresh water, and 3.47 kg/h of hydrogen. Furthermore, in the base case, the exergy efficiency, total cost rate, and unit cost of poly-generation are determined to be 23.9%, 9337 \$/h, and 19.42 \$/GJ, respectively. Additionally, the primary contributor to the total cost, $\dot{C}_{tot}$, is attributable to $\dot{C}_{dest,tot}$, which contributes 51.31% to the overall cost. In contrast, among the various cost rates, $\dot{C}_{env}$ has the lowest contribution percentages at approximately 3.95%. Additionally, the production of freshwater is considerably more expensive than other unit costs, with a value of 389.7 \$/GJ.

The sensitivity analysis of the system under study revealed that increasing the variables of N_{mirror} , x_{ch_4} and T_5 had a favorable effect on the system's exergy-economic and environmental metrics, while increasing GTIT had a negative

effect on the scheme's overall exergy-economic and environmental metrics. Moreover, the augmentation of GTIT resulted in an increase in hydrogen and freshwater production, as well as an improvement in poly-generation exergy efficiency due to the positive impact of R_p , T_5 and GTIT, while the opposite effect was observed from the upsurge in N_{mirror} and x_{ch_4} .

The utilization of the NSGA-II algorithm in the three-objective optimization of the examined layout yielded $\eta_{ex,poly}$, $\dot{C}_{tot}$, and c_{poly} values of 25.10%, 8482.67 \$/h, and 19.01 \$/GJ, respectively. These results exhibit improvements when compared to the outputs of the base case.

Appendix A

Table A Basic relationships for some configuration components

Correlation	Note	Number
Solar power tower [43, 52]		
$\dot{Q}_{use,rec} = \dot{m}_{air} (h_5 - h_6)$	SPT heat absorption	(A-1)
$\dot{Q}_{use,rec} = \alpha_{rec} \dot{Q}_{i,rec} - \dot{Q}_{loss,rec}$		(A-2)
$\dot{Q}_{i,rec} = DNI A_{mirror} \eta_{field}$	Inlet energy to receiver	(A-3)
$\dot{Q}_{loss,rec} = \dot{Q}_{loss,rad} + \dot{Q}_{loss,conv}$	Energy losses of the receiver	(A-4)
$\dot{Q}_{i,rec} = DNI A_{mirror} \eta_{field}$		(A-5)
$\dot{Q}_{loss,rec} = \dot{Q}_{loss,rad} + \dot{Q}_{loss,conv}$		(A-6)
$\dot{Q}_{loss,rad} = A_{rec} \cdot \epsilon m_{rec} \cdot \sigma \cdot T_{rec}^4$	Radiative heat losses	(A-7)
$\dot{Q}_{loss,conv} = A_{rec} \cdot \xi_{conv} \cdot (T_{sur,rec} - T_o)$	Convective heat losses	(A-8)
$T_{sur,rec} = T_{rec,e} + 20$	Temperature of receiver	(A-9)
$\xi_{conv} = (\xi_{force}^{3.2} + \xi_{natural}^{3.2})^{0.3125}, \xi_{force} = \frac{14 W}{m^2 K}, \xi_{natural} = \frac{9.09 W}{m^2 K}$	The losses related to convective heat transfer	(A-10)
Combustion chamber [53, 54]		
$\frac{\dot{n}_{fuel}}{\dot{n}_{air}} = \bar{\gamma} \rightarrow \frac{\dot{n}_{gas}}{\dot{n}_{air}} = 1 + \bar{\gamma}$	Molar rate relations	(A-11)
$\bar{\gamma} [x_{ch_4} CH_4 + (1 - x_{ch_4}) CO_2]$ $+ [0.7748 N_2 + 0.2059 O_2 + 0.003 CO_2 + 0.019 H_2 O]$ $\rightarrow [1 + \bar{\gamma}] [Y_{N_2} N_2 + Y_{O_2} O_2 + Y_{CO_2} CO_2 + Y_{H_2O} H_2 O]$	The chemical reaction between streams	(A-12)

$Y_{N_2} = \frac{0.7748}{1 + \bar{y}}$	Molar fraction of Nitrogen	(A-13)
$Y_{O_2} = \frac{0.2059 - 2 \cdot x_{ch4} \cdot \bar{y}}{1 + \bar{y}}$	Molar fraction of Oxygen	(A-14)
$Y_{CO_2} = \frac{0.003 + \bar{y}}{1 + \bar{y}}$	Molar fraction of Carbon dioxide	(A-15)
$Y_{H_2O} = \frac{0.019 + 2 \cdot x_{ch4} \cdot \bar{y}}{1 + \bar{y}}$	Molar fraction of Water	(A-16)
$-0.02 \bar{y} \overline{LHV} + \bar{h}_{air} + \bar{y} \bar{h}_{fuel} - (1 + \bar{y}) \bar{h}_{gas} = 0$	The energy equation	(A-17)
Thermoelectric generator [46, 55]		
$\dot{W}_{TEG} = \eta_{TEG} \cdot \dot{Q}_{coldside}$	Output power	(A-18)
$\dot{Q}_{coldside} = \dot{m}_{cold} (h_{out} - h_{in})$	Cold side energy rate	(A-19)
$\eta_{TEG} = \eta_{Carnot} \frac{\sqrt{1 + ZT_M} - 1}{\sqrt{1 + ZT_M} + \frac{T_L}{T_H}}$	Efficiency of TEG	(A-20)
$\eta_{Carnot} = 1 - \frac{T_L}{T_H}$	Carnot efficiency	(A-21)
Alkaline electrolyzer [47, 48]		
$\dot{n}_{hydrogen} = \frac{J}{2F}$	Molar flow rate of produced hydrogen	(A-22)
$\dot{W}_{SBC\ Turbine} + \dot{W}_{TEG} - \dot{W}_{SBC\ Compressor} = \dot{W}_{AE} = JV$	Input power to AE	(A-23)
$V = V_0 + V_{ohm} + V_{act,ca} + V_{act,an}$	Total overvoltage	(A-24)
$V_0 = 1.229 - 0.00085(T_{AE} - 298)$	Nernst voltage	(A-25)
$V = V_0 + [(r_1 + d_1) + r_2 T_{AE} + d_2 P_{AE}]J + s \log[(t_1 + t_2/T_{AE} + t_3/T_{AE}^2)J + 1]$	Total overvoltage	(A-26)
Ejector [35, 36, 56, 57] (see Fig. 2)		
$V_N = \left(\eta_N \cdot 2 \cdot \frac{1000 \cdot (P_{con} - P_{eva})}{\rho_{34}} \right)^{0.5}$	Nozzle velocity	(A-27)
$A_N = \pi \cdot \frac{D_N^2}{4}$	Nozzle area	(A-28)
$\rho_{34} = f(T_{34}, x_{34})$	Density of poor solution	(A-29)
$A_N = \frac{\dot{m}_{33}}{\rho_{34} \cdot V_N}$	Refer to Eq. (A-27)	(A-30)
$V_M = \frac{\dot{m}_{28} + \dot{m}_{34}}{A_M \cdot \rho_M}$	Mixing tube velocity	(A-31)
$A_M = \pi \cdot \frac{D_M^2}{4}$	Mixing tube Area	(A-32)

Mixing tube	$\rho_M = \frac{\dot{m}_{28} + \dot{m}_{34}}{\frac{\dot{m}_{28}}{\rho_{28}} + \frac{\dot{m}_{34}}{\rho_{34}}}$	Mixing tube stream density	(A-33)
	$P_M = P_{eva} + \eta_M \cdot \frac{\dot{m}_{34} \cdot V_N - (\dot{m}_{34} + \dot{m}_{28}) \cdot V_M}{A_M \cdot 1000}$	Mixing tube stream pressure	(A-34)
	$P_D = P_M + \frac{0.5 \cdot \rho_M \cdot (V_M^2 - V_D^2) \cdot \eta_D}{1000}$	Diffuser pressure	(A-35)
	$P_{abs} = P_D$		(A-36)
	$V_D = \frac{(\dot{m}_{34} + \dot{m}_{28})}{\rho_M \cdot A_D}$	Diffuser velocity	(A-37)
Diffuser	$A_D = \pi \cdot \frac{D_D^2}{4}$	Diffuser area	(A-38)
	$\dot{m}_{34} \cdot h_{34} + \dot{m}_{28} \cdot h_{28} = \dot{m}_{34} \cdot h_{35s} + \dot{m}_{28} \cdot h_{35r}$	Ejector energy balance	(A-39)
	$h_{35s} = f(T_{35}, X_{35})$	Enthalpy of weak solution (state 35)	(A-40)
	$h_{35r} = f(T_{35}, P_{ab})$	Enthalpy of refrigerant vapor (state 35)	(A-41)
MED [33, 34, 58, 59]			
	$F_T = D_T \left(\frac{1}{1 - \frac{X_{SW}}{X_n}} \right)$	The total feed water to effects	(A-42)
	$B_n = F_T - D_T$	The total brine	(A-43)
	$F_i = \frac{F_T}{N_E}, i = 1:N_E$	The feed water for each effect	(A-44)
	$\Delta T_E = \frac{T_{HS,MED} - T_F}{N_E + 1}$	Temperature difference between effects	(A-45)
	$T_i = T_{i+1} + \Delta T_E, i = (N_E - 1):1$	The boiling temperature in each effect	(A-46)
	$B_1 = F_1 - D_1$	Mass balance of first effect	(A-47)
	$X_1 \cdot B_1 = X_{SW} \cdot F_1$	Salt balance of first effect	(A-48)
	$\dot{m}_{20} \cdot (h_{20} - h_{21}) = F_1 \cdot C_{P_w} \cdot (T_1 - T_F) + D_1 \cdot LH(T_{V1})$	Energy balance of first effect	(A-49)
	$A_1 = \frac{Q_{E1}}{U_{E1} \cdot (T_{HS,MED} - T_{first, effect})}, Q_{E1} = \dot{m}_{20} \cdot (h_{20} - h_{21})$	Area of heat transfer for the first effect	(A-50)
	$B_i = F_i - D_i + B_{i-1}$	Mass balance of other effects	(A-51)

$B_i \cdot X_i = B_{i-1} \cdot X_{i-1} + X_F \cdot F_i$	Salt balance of other effects	(A-52)
$D_{i-1} \cdot LH(T_{V_1}) + B_{i-1} \cdot C_{P_w} \cdot (T_{i-1} - T_i)$ $= D_i \cdot LH(T_{V_1}) + F_i \cdot C_{P_w} \cdot (T_i - T_f)$	Energy balance of other effects	(A-53)
$A_i = \frac{Q_{E_i}}{U_{E_i} \cdot (T_{V_{i-1}} - T_i)}, Q_{E_i} = D_{i-1} \cdot LH(T_{V_i})$	Area of heat transfer for the other effects	(A-54)
$A_c = \frac{Q_{C_{MED}}}{U_{C_{MED}} \cdot LMTD_{C_{MED}}}, Q_{C_{MED}} = D_n \cdot LH(T_{V_n})$	Area of heat transfer the MED condenser	(A-55)
$LMTD_{C_{MED}} = \frac{(T_{V_n} - T_F) - (T_{V_n} - T_{SW})}{\ln(\frac{T_{V_n} - T_F}{T_{V_n} - T_{SW}})}$	The condenser logarithmic mean temperature difference	(A-56)
$A_{MED} = \sum_{i=1}^{N_E} A_{E_i} + A_c$	Total heat transfer Area of MED unit	(A-57)
$GOR = D_T / \dot{m}_{20}$	Gain Ratio	(A-58)

Appendix B

Table B Cost estimation formulas

Device	Equation	Reference
Compressor 1	$Z_{com,1} = 71.1 \dot{m}_{air} \left(\frac{1}{0.9 - \eta_{air,com}} \right) \left(\frac{P_2}{P_1} \right) \ln \left(\frac{P_2}{P_1} \right)$	[60]
HWH	$Z_{HWH} = 130 \left(\frac{A_{HWH}}{0.093} \right)^{0.78}$	[61]
Compressor 2	$Z_{com,2} = 71.1 \dot{m}_{air} \left(\frac{1}{0.9 - \eta_{air,com}} \right) \left(\frac{P_4}{P_3} \right) \ln \left(\frac{P_4}{P_3} \right)$	[60]
GTC Recuperator	$Z_{GTC,Rec} = 4122 A_{Rec}^{0.6}$	[37]
SPT	$Z_{SPT} = 170 A_{mirror} + 3 \times 10^6 \exp \left(0.0113 \times \left(\frac{H_{tower} - L_{rec}}{2} + \frac{L_{mirror}}{2} \right) \right)$ $+ 5.54028 \times 10^7 \left(\frac{A_{rec}}{1110} \right)^{0.7}$	[60]
Combustion chamber	$Z_{cc} = \frac{48.08 \dot{m}_{air}}{0.995 - \frac{P_7}{6}} (1 + \exp(0.018 T_7 - 26.4))$	[60]
GTC Turbine	$Z_{GTC,tur} = 479.34 \dot{m}_{gas} \left(\frac{1}{0.92 - \eta_{GTC,tur}} \right) \ln \left(\frac{P_7}{P_8} \right) (1$ $+ \exp(0.036 T_7 - 54.4))$	[60]

RHX	$Z_{RHX} = 4122 A_{RHX}^{0.6}$	[37]
SBC Turbine	$Z_{SBC,tur} = 479.34 \dot{m}_{SBC} \left(\frac{1}{0.92 - \eta_{SBC,tur}} \right) \ln \left(\frac{P_{13}}{P_{14}} \right) (1 + \exp(0.036T_{13} - 54.4))$	[62]
SBC Compressor	$Z_{SBC,com} = 71.1 \dot{m}_{SBC} \left(\frac{1}{0.9 - \eta_{SBC,com}} \right) \left(\frac{P_{17}}{P_{16a}} \right) \ln \left(\frac{P_{17}}{P_{16a}} \right)$	[62]
SBC Recuperator	$Z_{SBC,Rec} = 4122 A_{Rec}^{0.6}$	[37]
TEG	$Z_{TEG} = 1500 \dot{W}_{TEG}$	[63]
HRSG	$Z_{HRSG} = 6570 \sum \left(\frac{\dot{Q}_j}{\Delta T_{lmt,d,j}} \right)^{0.8} + 21276 \dot{m}_{20} + 1184.4(\dot{m}_{gas})^{1.2}$	[39]
MED effects	$Z_{MED} = 201.67 \cdot Q \cdot \Delta T_{LMTD}^{-1}$	[34]
MED condenser	$Z_{MED,con} = 430.0582 \cdot Q \cdot \Delta T_{LMTD}^{-1}$	[34]
ACH generator	$Z_{ACH,gen} = Z_{R,gen} \left(\frac{A_{ACH,gen}}{A_R} \right), Z_{R,gen} = 17500 \$, A_R = 100 m^2$	[32]
ACH absorber	$Z_{ACH,abs} = Z_{R,abs} \left(\frac{A_{ACH,abs}}{A_R} \right), Z_{R,abs} = 16500 \$, A_R = 100 m^2$	[32]
ACH evaporator	$Z_{ACH,eva} = Z_{R,eva} \left(\frac{A_{ACH,eva}}{A_R} \right), Z_{R,eva} = 16000 \$, A_R = 100 m^2$	[32]
ACH condenser	$Z_{ACH,con} = Z_{R,con} \left(\frac{A_{ACH,con}}{A_R} \right), Z_{R,con} = 8000 \$, A_R = 100 m^2$	[32]
ACH SHX	$Z_{ACH,SHX} = Z_{R,SHX} \left(\frac{A_{ACH,SHX}}{A_R} \right), Z_{R,SHX} = 12000 \$, A_R = 100 m^2$	[32]
ACH pump	$Z_{ACH,pump} = Z_{R,pump} \left(\frac{\dot{W}_{ACH,pump}}{\dot{W}_{R,pump}} \right)^{0.26} \left(\frac{1 - \eta_{pump}}{\eta_{pump}} \right)^{0.5}, Z_{R,pump} = 2100 \$, \dot{W}_{R,pump} = 10 kW$	[32]
ACH Ejector	$Z_{Ejector} = 1000 \times 16.14 \times 0.989 \times \left(\dot{m}_{34} \left(\frac{T_{34}}{(0.001P_{34})} \right)^{0.05} P_{35}^{-0.75} \right)$	[64]
ACH EV	$Z_{ACH,EV} = 114.5 \dot{m}_{26}$	[32]
AE	$Z_{AE} = 400 \dot{W}_{AE}$	[47]

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Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: